

Advance

2Gb C-die NAND Flash

Single-Level-Cell (1bit/cell)

datasheet

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1.0 INTRODUCTION

1.1 Product List

Part Number	Vcc Range	Organization	PKG Type
K9F2G08U0C-S	2.7 ~ 3.6v	x8	TSOP1
K9F2G08U0C-H	2.7 ~ 3.6v	x8	63 FBGA

1.2 Features

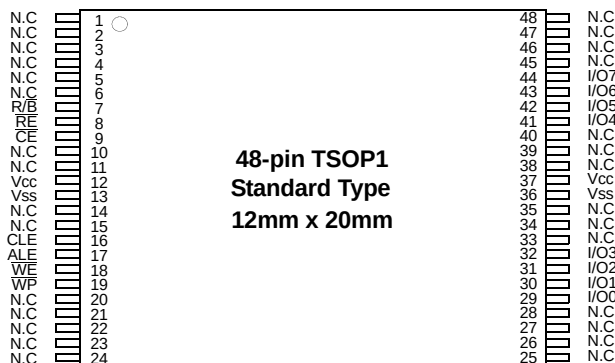
- Voltage Supply
 - 3.3V device(K9F2G08U0C): 2.70V ~ 3.60V
- Organization
 - Memory Cell Array : (256M + 8M) x 8bit
 - Data Register : (2K + 64) x 8bit
- Automatic Program and Erase
 - Page Program : (2K + 64)Byte
 - Block Erase : (128K + 4K)Byte
- Page Read Operation
 - Page Size : (2K + 64)Byte
 - Random Read : 40μs(Max.)
 - Serial Access : 25ns(Min.)
- Fast Write Cycle Time
 - Page Program time : 250μs(Typ.)
 - Block Erase Time : 2ms(Typ.)
- Command/Address/Data Multiplexed I/O Port
- Hardware Data Protection
 - Program/Erase Lockout During Power Transitions
- Reliable CMOS Floating-Gate Technology
 - Endurance & Data Retention : Refor to the gualification report
 - ECC regnirement : 1 bit / 528bytes• Command Driven Operation
- Unique ID for Copyright Protection
- Package :
 - K9F2G08U0C-SCB0/SIB0 : Pb-FREE PACKAGE
48 - Pin TSOP I (12 x 20 / 0.5 mm pitch)
 - K9F2G08U0C-HCB0/HIB0 : Pb-FREE PACKAGE
63 - ball FBGA (9 x 11 / 0.8 mm pitch)

1.3 General Description

Offered in 256Mx8bit, the K9F2G08U0C is a 2G-bit NAND Flash Memory with spare 64M-bit. Its NAND cell provides the most cost-effective solution for the solid state application market. A program operation can be performed in typical 250μs on the (2K+64)Byte page and an erase operation can be performed in typical 2ms on a (128K+4K)Byte block. Data in the data register can be read out at 25ns cycle time per Byte. The I/O pins serve as the ports for address and data input/output as well as command input. The on-chip write controller automates all program and erase functions including pulse repetition, where required, and internal verification and margining of data. Even the write-intensive systems can take advantage of the K9F2G08U0C's extended reliability of 100K program/erase cycles by providing ECC(Error Correcting Code) with real time mapping-out algorithm. The K9F2G08U0C is an optimum solution for large nonvolatile storage applications such as solid state file storage and other portable applications requiring non-volatility.

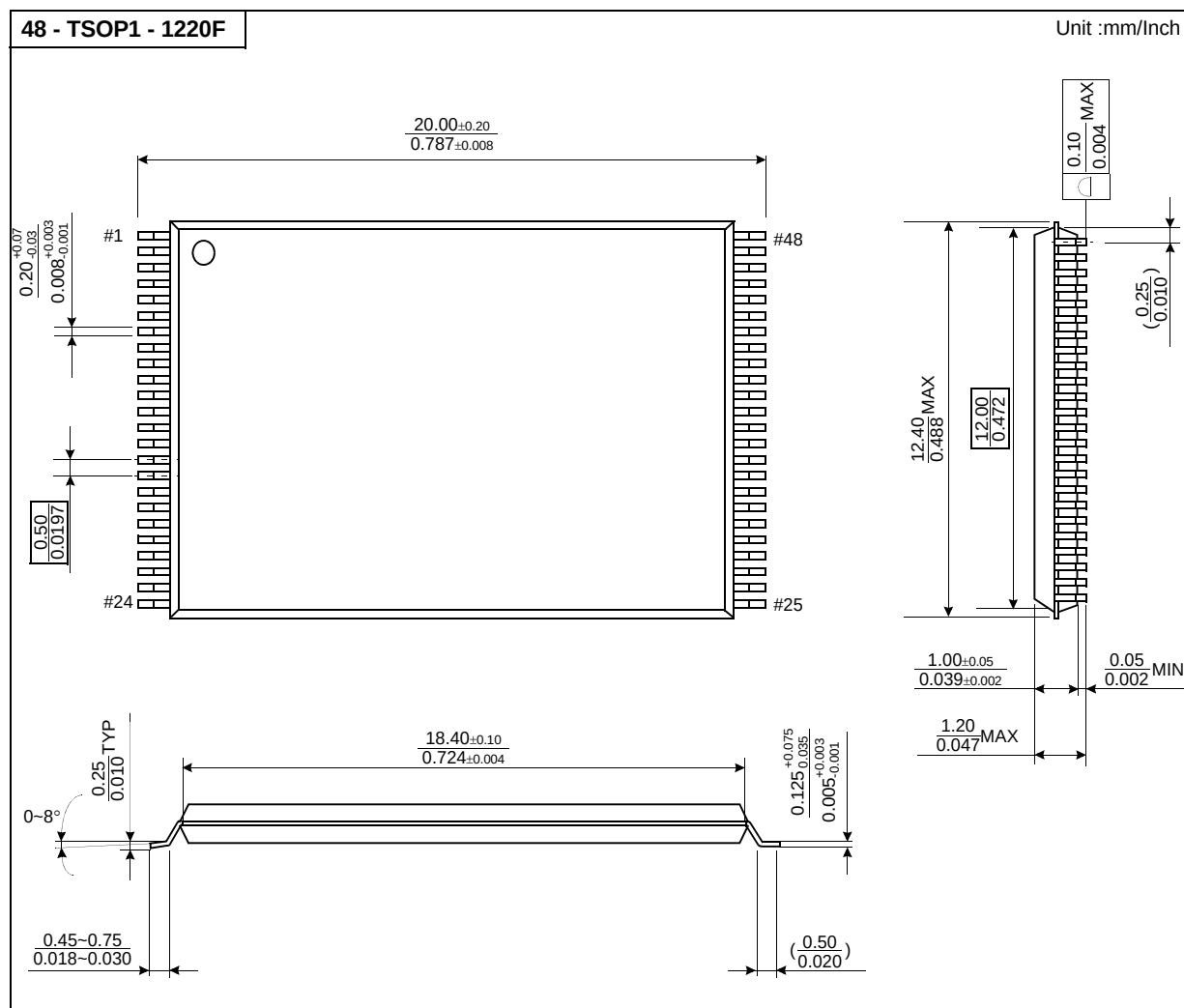
1.4 Pin Configuration (TSOP1)

K9F2G08U0C-SCB0/SIB0

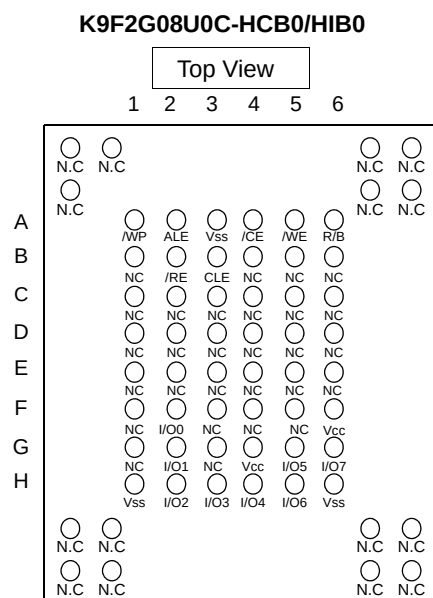


1.4.1 Package Dimensions

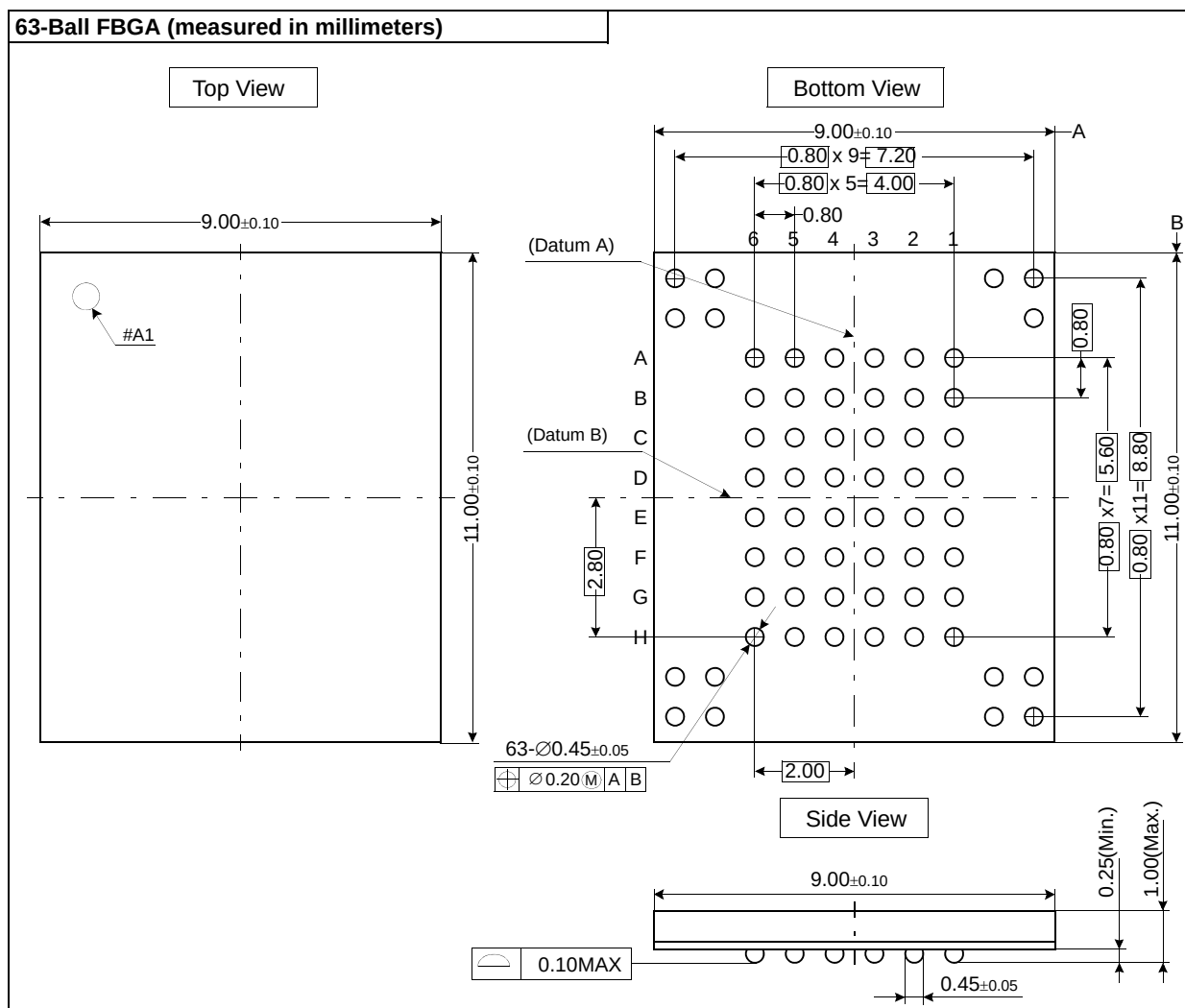
48-PIN LEAD/LEAD FREE PLASTIC THIN SMALL OUT-LINE PACKAGE TYPE(I)



1.5 Pin Configuration (FBGA)



1.5.1 Package Dimensions



1.6 Pin Description

Pin Name	Pin Function
I/O ₀ ~ I/O ₇	DATA INPUTS/OUTPUTS The I/O pins are used to input command, address and data, and to output data during read operations. The I/O pins float to high-z when the chip is deselected or when the outputs are disabled.
CLE	COMMAND LATCH ENABLE The CLE input controls the activating path for commands sent to the command register. When active high, commands are latched into the command register through the I/O ports on the rising edge of the $\overline{WE}$ signal.
ALE	ADDRESS LATCH ENABLE The ALE input controls the activating path for address to the internal address registers. Addresses are latched on the rising edge of $\overline{WE}$ with ALE high.
$\overline{CE}$	CHIP ENABLE The $\overline{CE}$ input is the device selection control. When the device is in the Busy state, $\overline{CE}$ high is ignored, and the device does not return to standby mode in program or erase operation.
$\overline{RE}$	READ ENABLE The $\overline{RE}$ input is the serial data-out control, and when active drives the data onto the I/O bus. Data is valid t _{REA} after the falling edge of $\overline{RE}$ which also increments the internal column address counter by one.
$\overline{WE}$	WRITE ENABLE The $\overline{WE}$ input controls writes to the I/O port. Commands, address and data are latched on the rising edge of the $\overline{WE}$ pulse.
$\overline{WP}$	WRITE PROTECT The $\overline{WP}$ pin provides inadvertent program/erase protection during power transitions. The internal high voltage generator is reset when the $\overline{WP}$ pin is active low.
R/ $\overline{B}$	READY/BUSY OUTPUT The R/ $\overline{B}$ output indicates the status of the device operation. When low, it indicates that a program, erase or random read operation is in process and returns to high state upon completion. It is an open drain output and does not float to high-z condition when the chip is deselected or when outputs are disabled.
V _{cc}	POWER V _{cc} is the power supply for device.
V _{ss}	GROUND
N.C	NO CONNECTION Lead is not internally connected.

NOTE :

Connect all V_{CC} and V_{SS} pins of each device to common power supply outputs.
Do not leave V_{CC} or V_{SS} disconnected.

2.0 PRODUCT INTRODUCTION

NAND Flash Memory has addresses multiplexed into 8 I/Os. This scheme dramatically reduces pin counts and allows system upgrades to future densities by maintaining consistency in system board design. Command, address and data are all written through I/O's by bringing $\overline{WE}$ to low while $\overline{CE}$ is low. Those are latched on the rising edge of $\overline{WE}$. Command Latch Enable(CLE) and Address Latch Enable(ALE) are used to multiplex command and address respectively, via the I/O pins. Some commands require one bus cycle. For example, Reset Command, Status Read Command, etc. require just one cycle bus. Some other commands, like page read and block erase and page program, require two cycles: one cycle for setup and the other cycle for execution. Page Read and Page Program need the same five address cycles following the required command input. In Block Erase operation, however, only the three row address cycles are used. Device operations are selected by writing specific commands into the command register. Table 1 defines the specific commands of the K9G2G08U0C.

[Table 1] Command Sets 支持的命令字集

Function	1st Cycle	2nd Cycle	Acceptable Command during Busy
Read	00h	30h	
Read for Copy Back	00h	35h	
Read ID	90h	-	
Reset	FFh	-	O
Page Program	80h	10h	
Copy-Back Program	85h	10h	
Two-Plane Page Program ⁽²⁾	80h---11h	81h---10h	
Block Erase	60h	D0h	
Random Data Input ⁽¹⁾	85h	-	
Random Data Output ⁽¹⁾	05h	E0h	
Read Status	70h	-	O
Read Status 2	F1h	-	O

NOTE :

- 1) Random Data Input/Output can be executed in a page.
- 2) Any command between 11h and 81h is prohibited except 70h/F1h and FFh.

Caution :

Any undefined command inputs are prohibited except for above command set of Table 1.

2.1 Absolute Maximum Ratings

Parameter		Symbol	Rating	Unit
Voltage on any pin relative to VSS		V _{CC}	-0.6 to +4.6	V
		V _{IN}	-0.6 to +4.6	
		V _{I/O}	-0.6 to V _{CC} + 0.3 (< 4.6V)	
Temperature Under Bias	K9F2G08U0C-XCB0	T _{BIAS}	-10 to +125	°C
	K9F2G08U0C-XIB0		-40 to +125	
Storage Temperature	K9F2G08U0C-XCB0	T _{STG}	-65 to +150	°C
	K9F2G08U0C-XIB0			
Short Circuit Current		I _{OS}	5	mA

NOTE :

- 1) Minimum DC voltage is -0.6V on input/output pins. During transitions, this level may undershoot to -2.0V for periods <30ns.
Maximum DC voltage on input/output pins is V_{CC}+0.3V which, during transitions, may overshoot to V_{CC}+2.0V for periods <20ns.
- 2) Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to the conditions as detailed in the operational sections of this data sheet. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

2.2 Recommended Operating Conditions

(Voltage reference to GND, K9F2G08U0C-XCB0 :T_A=0 to 70°C, K9F2G08U0C-XIB0:T_A=-40 to 85°C)

Parameter	Symbol	3.3V			Unit
		Min	Typ.	Max	
Supply Voltage	V _{CC}	2.7	3.3	3.6	V
Supply Voltage	V _{SS}	0	0	0	V

2.3 DC And Operating Characteristics(Recommended operating conditions otherwise noted.)

Parameter		Symbol	Test Conditions	3.3V			Unit
				Min	Typ	Max	
Operating Current	Page Read with Serial Access	I _{CC1}	t _{RC} =25ns CE=V _{IL} , I _{OUT} =0mA	-	20	35	mA
	Program	I _{CC2}	-				
	Erase	I _{CC3}	-				
Stand-by Current(TTL)		I _{SB1}	CE=V _{IH} , WP=0V/V _{CC}	-	-	1	μA
Stand-by Current(CMOS)		I _{SB2}	CE=V _{CC} -0.2, WP=0V/V _{CC}	-	10	80	
Input Leakage Current		I _{LI}	V _{IN} =0 to V _{CC} (max)	-	-	±10	
Output Leakage Current		I _{LO}	V _{OUT} =0 to V _{CC} (max)	-	-	±10	V
Input High Voltage		V _{IH} (1)	-	0.8xV _{CC}	-	V _{CC} +0.3	
Input Low Voltage, All inputs		V _{IL} (1)	-	-0.3	-	0.2xV _{CC}	
Output High Voltage Level		V _{OH}	K9F2G08B0C: I _{OH} =-100μA K9F2G08U0C: I _{OH} =-400μA	2.4	-	-	V
Output Low Voltage Level		V _{OL}	K9F2G08B0C: I _{OL} =100μA K9F2G08U0C: I _{OL} =2.1mA	-	-	0.4	
Output Low Current(R/B)		I _{OL} (R/B)	K9F2G08B0C: V _{OL} =0.1V K9F2G08U0C: V _{OL} =0.4V	8	10	-	mA

NOTE :

- 1) V_{IL} can undershoot to -0.4V and V_{IH} can overshoot to V_{CC} +0.4V for durations of 20 ns or less.
2) Typical value is measured at V_{CC}= 3.3V, T_A=25°C. Not 100% tested.

2.4 Valid Block

Parameter	Symbol	Min	Typ.	Max	Unit
K9F2G08U0C	NvB	2,008	-	2,048	Blocks

NOTE :

- 1) The device may include initial invalid blocks when first shipped. Additional invalid blocks may develop while being used. The number of valid blocks is presented with both cases of invalid blocks considered. Invalid blocks are defined as blocks that contain one or more bad bits. Do not erase or program factory-marked bad blocks. Refer to the attached technical notes for appropriate management of invalid blocks.
- 2) The 1st block, which is placed on 00h block address, is guaranteed to be a valid block up to 1K program/erase cycles with 1bit/528Byte ECC.
- 3) The number of valid block is on the basis of single plane operations, and this may be decreased with two plane operations.

2.5 AC Test Condition

(K9F2G08U0C-XCB0 :TA=0 to 70°C, K9F2G08U0C-XIB0:TA=-40 to 85°C, K9F2G08U0C: Vcc=2.7V~3.6V unless otherwise noted)

Parameter	K9F2G08U0C
Input Pulse Levels	0V to Vcc
Input Rise and Fall Times	5ns
Input and Output Timing Levels	Vcc/2
Output Load	1 TTL GATE and CL=50pF

2.6 Capacitance(T_A=25°C, V_{CC}= 3.3V, f=1.0MHz)

Item	Symbol	Test Condition	Min	Max	Unit
Input/Output Capacitance	C _{I/O}	V _{IL} =0V	-	10	pF
Input Capacitance	C _{IN}	V _{IN} =0V	-	10	pF

NOTE :

Capacitance is periodically sampled and not 100% tested.

2.7 Mode Selection 各模式下电平情况说明

CLE	ALE	CE	WE	RE	WP	Mode	
H	L	L		H	X	Read Mode	Command Input
L	H	L		H	X		Address Input(5clock)
H	L	L		H	H	Write Mode	Command Input
L	H	L		H	H		Address Input(5clock)
L	L	L		H	H	Data Input	
L	L	L	H		X	Data Output	
X	X	X	X	H	X	During Read(Busy)	
X	X	X	X	X	H	During Program(Busy)	
X	X	X	X	X	H	During Erase(Busy)	
X	X ⁽¹⁾	X	X	X	L	Write Protect	
X	X	H	X	X	0V/Vcc ⁽²⁾	Stand-by	

NOTE :

1) X can be V_{IL} or V_{IH}.

2) WP should be biased to CMOS high or CMOS low for standby.

2.8 Program / Erase Characteristics 时序要求

Parameter	Symbol	Min	Typ	Max	Unit
Program Time	tPROG	-	250	750	μs
Dummy Busy time for Two-Plane Page Program	tDBSY	-	2.5	3	μs
Number of Partial Program Cycles	Nop	-	-	4	cycles
Block Erase Time	tBERS	-	2	10	ms

NOTE :

1) Typical value is measured at Vcc=3.3V, TA=25°C. Not 100% tested.

2) Typical program time is defined as the time within which more than 50% of the whole pages are programmed at 3.3V Vcc and 25°C temperature.

2.9 AC Timing Characteristics for Command / Address / Data Input

Parameter	Symbol	Min	Max	Unit
CLE Setup Time	tCLS ⁽¹⁾	12	-	ns
CLE Hold Time	tCLH	5	-	ns
CE Setup Time	tCS ⁽¹⁾	20	-	ns
CE Hold Time	tCH	5	-	ns
$\overline{\text{WE}}$ Pulse Width	tWP	12	-	ns
ALE Setup Time	tALS ⁽¹⁾	12	-	ns
ALE Hold Time	tALH	5	-	ns
Data Setup Time	tDS ⁽¹⁾	12	-	ns
Data Hold Time	tDH	5	-	ns
Write Cycle Time	tWC	25	-	ns
$\overline{\text{WE}}$ High Hold Time	tWH	10	-	ns
Address to Data Loading Time	tADL ⁽²⁾	100	-	ns

NOTE :

1) The transition of the corresponding control pins must occur only once while $\overline{\text{WE}}$ is held low2) tADL is the time from the $\overline{\text{WE}}$ rising edge of final address cycle to the $\overline{\text{WE}}$ rising edge of first data cycle

2.10 AC Characteristics for Operation

Parameter	Symbol	Min	Max	Unit
Data Transfer from Cell to Register	t _{tr}	-	40	μs
ALE to $\overline{RE}$ Delay	t _{AR}	10	-	ns
CLE to $\overline{RE}$ Delay	t _{CLR}	10	-	ns
Ready to $\overline{RE}$ Low	t _{RR}	20	-	ns
$\overline{RE}$ Pulse Width	t _{RP}	12	-	ns
$\overline{WE}$ High to Busy	t _{WB}	-	100	ns
Read Cycle Time	t _{RC}	25	-	ns
$\overline{RE}$ Access Time	t _{REA}	-	20	ns
$\overline{CE}$ Access Time	t _{CEA}	-	25	ns
$\overline{RE}$ High to Output Hi-Z	t _{RHZ}	-	100	ns
$\overline{CE}$ High to Output Hi-Z	t _{CHZ}	-	30	ns
$\overline{CE}$ High to ALE or CLE Don't Care	t _{CSD}	10	-	ns
$\overline{RE}$ High to Output Hold	t _{RHOH}	15	-	ns
$\overline{CE}$ High to Output Hold	t _{COH}	15	-	ns
$\overline{RE}$ High Hold Time	t _{REH}	15	-	ns
Output Hi-Z to $\overline{RE}$ Low	t _{IR}	0	-	ns
$\overline{RE}$ High to $\overline{WE}$ Low	t _{RHW}	100	-	ns
$\overline{WE}$ High to $\overline{RE}$ Low	t _{WHR}	60	-	ns
Device Resetting Time(Read/Program/Erase)	t _{RST}	-	5/10/500 ⁽¹⁾	μs

NOTE :

1) If reset command(FFh) is written at Ready state, the device goes into Busy for maximum 5μs.

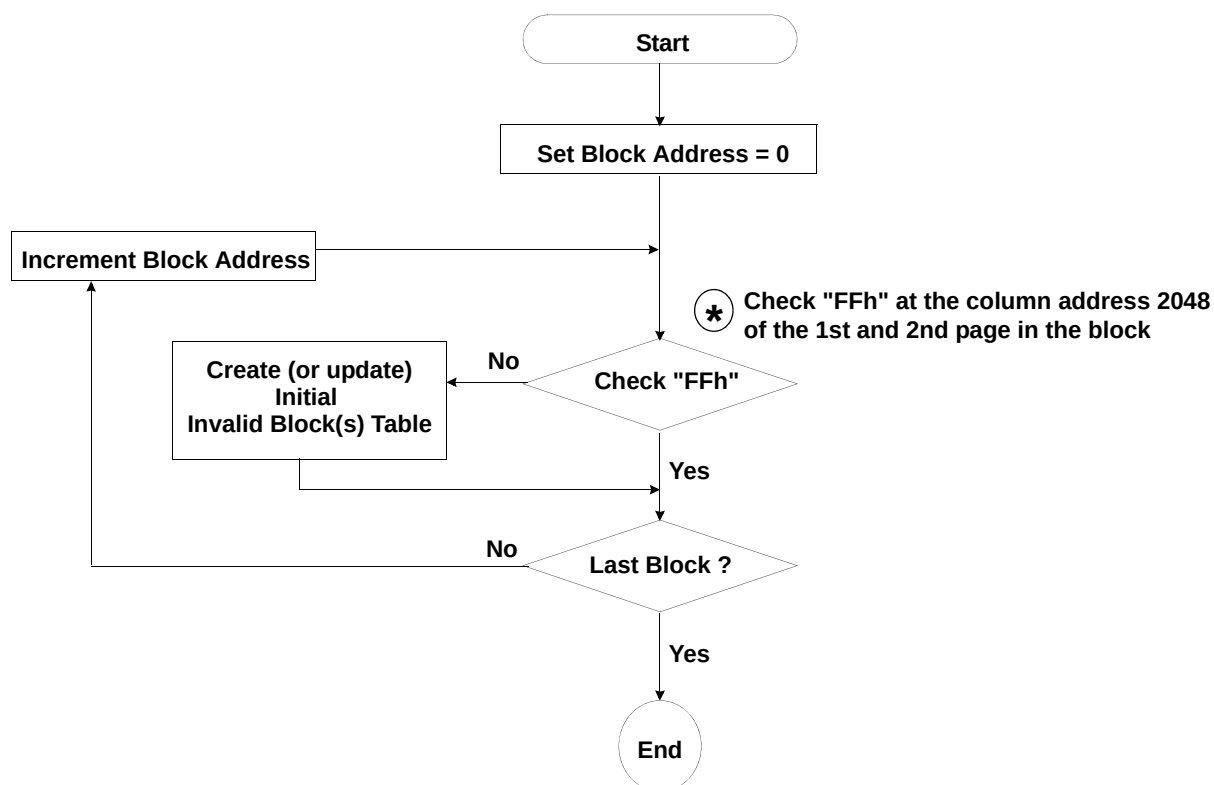
3.0 NAND FLASH TECHNICAL NOTES

3.1 Initial Invalid Block(s)

Initial invalid blocks are defined as blocks that contain one or more initial invalid bits whose reliability is not guaranteed by Samsung. The information regarding the initial invalid block(s) is called the initial invalid block information. Devices with initial invalid block(s) have the same quality level as devices with all valid blocks and have the same AC and DC characteristics. An initial invalid block(s) does not affect the performance of valid block(s) because it is isolated from the bit line and the common source line by a select transistor. The system design must be able to mask out the initial invalid block(s) via address mapping. The 1st block, which is placed on 00h block address, is guaranteed to be a valid block up to 1K program/erase cycles with 1bit/528Byte ECC.

3.2 Identifying Initial Invalid Block(s)

All device locations are erased(FFh) except locations where the initial invalid block(s) information is written prior to shipping. The initial invalid block(s) status is defined by the 1st byte in the spare area. Samsung makes sure that either the 1st or 2nd page of every initial invalid block has non-FFh data at the column address of 2048. Since the initial invalid block information is also erasable in most cases, it is impossible to recover the information once it has been erased. Therefore, the system must be able to recognize the initial invalid block(s) based on the original initial invalid block information and create the initial invalid block table via the following suggested flow chart(Figure 1). Any intentional erasure of the original initial invalid block information is prohibited.



[Figure 1] Flow chart to create initial invalid block table

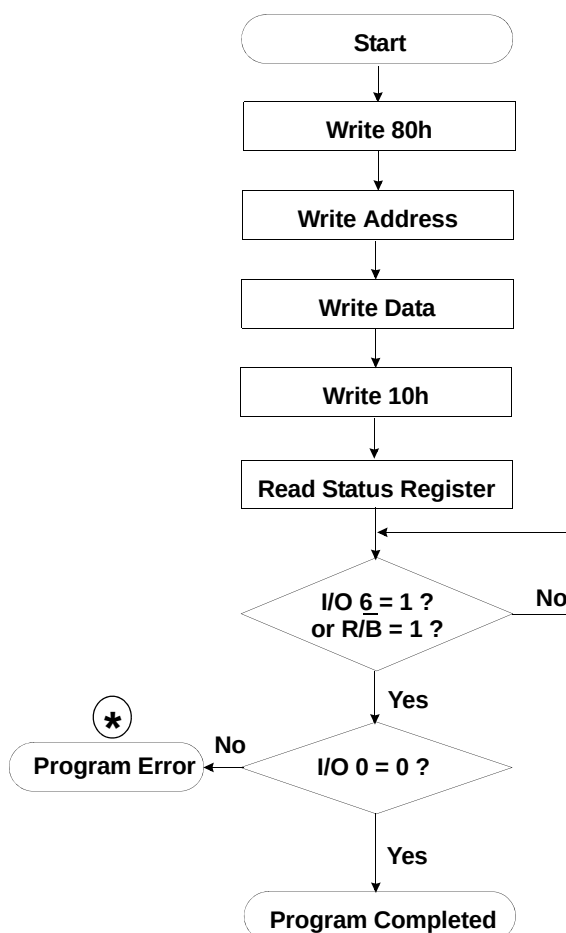
NAND Flash Technical Notes (Continued)

3.3 Error In Write Or Read Operation

Within its life time, additional invalid blocks may develop with NAND Flash memory. Refer to the qualification report for the actual data. The following possible failure modes should be considered to implement a highly reliable system. In the case of status read failure after erase or program, block replacement should be done. Because program status fail during a page program does not affect the data of the other pages in the same block, block replacement can be executed with a page-sized buffer by finding an erased empty block and reprogramming the current target data and copying the rest of the replaced block. In case of Read, ECC must be employed. To improve the efficiency of memory space, it is recommended that the read or verification failure due to single bit error be reclaimed by ECC without any block replacement. The said additional block failure rate does not include those reclaimed blocks.

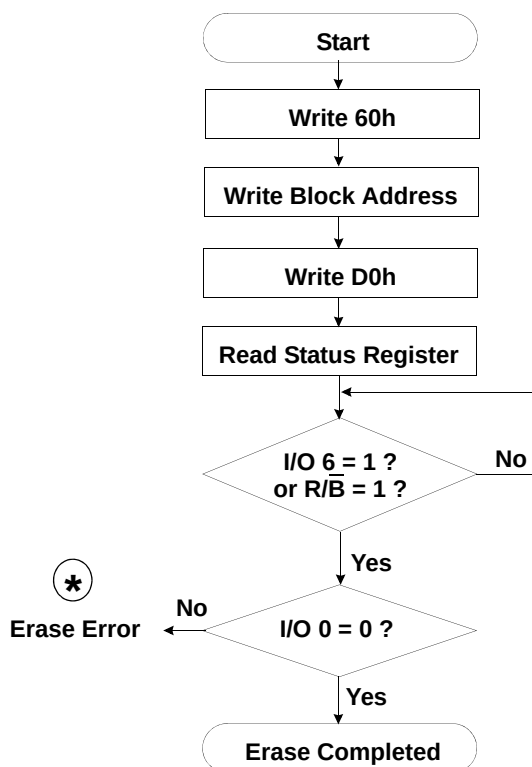
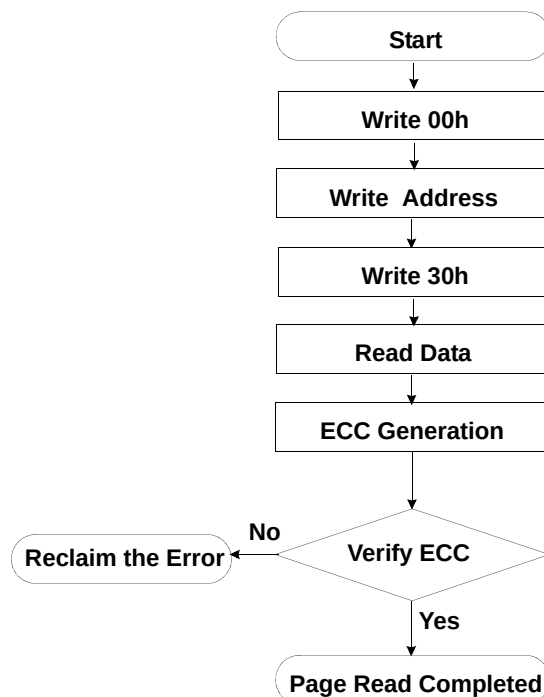
Failure Mode		Detection and Countermeasure sequence
Write	Erase Failure	Status Read after Erase --> Block Replacement
	Program Failure	Status Read after Program --> Block Replacement
Read	Single Bit Failure	Verify ECC -> ECC Correction

ECC : Error Correcting Code --> Hamming Code etc.
Example) 1bit correction & 2bit detection

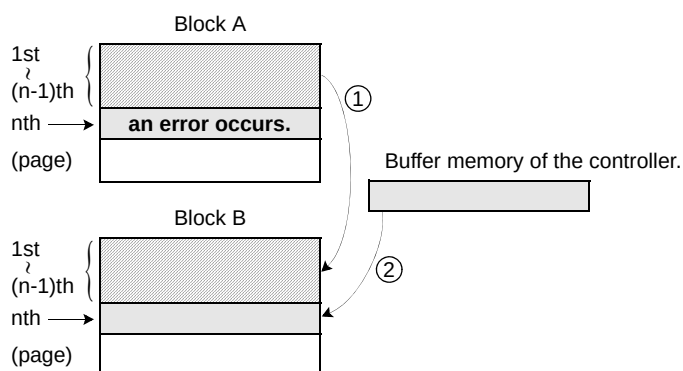
Program Flow Chart

***** : If program operation results in an error, map out the block including the page in error and copy the target data to another block.

NAND Flash Technical Notes (Continued)

Erase Flow ChartRead Flow Chart

(*) : If erase operation results in an error, map out the failing block and replace it with another block.

Block Replacement

* Step1

When an error happens in the nth page of the Block 'A' during erase or program operation.

* Step2

Copy the data in the 1st ~ (n-1)th page to the same location of another free block. (Block 'B')

* Step3

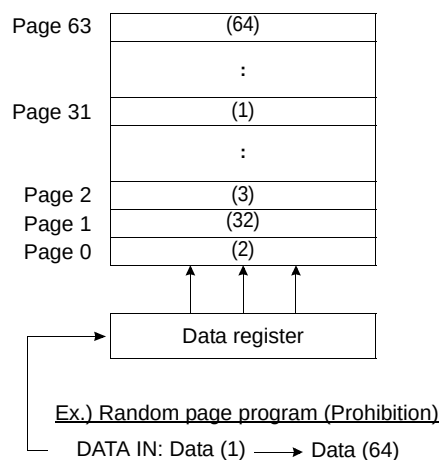
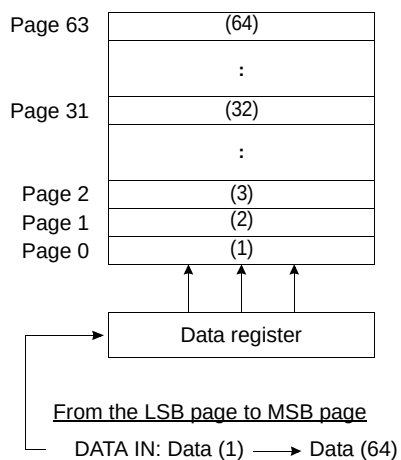
Then, copy the nth page data of the Block 'A' in the buffer memory to the nth page of the Block 'B'.

* Step4

Do not erase or program to Block 'A' by creating an 'invalid block' table or other appropriate scheme.

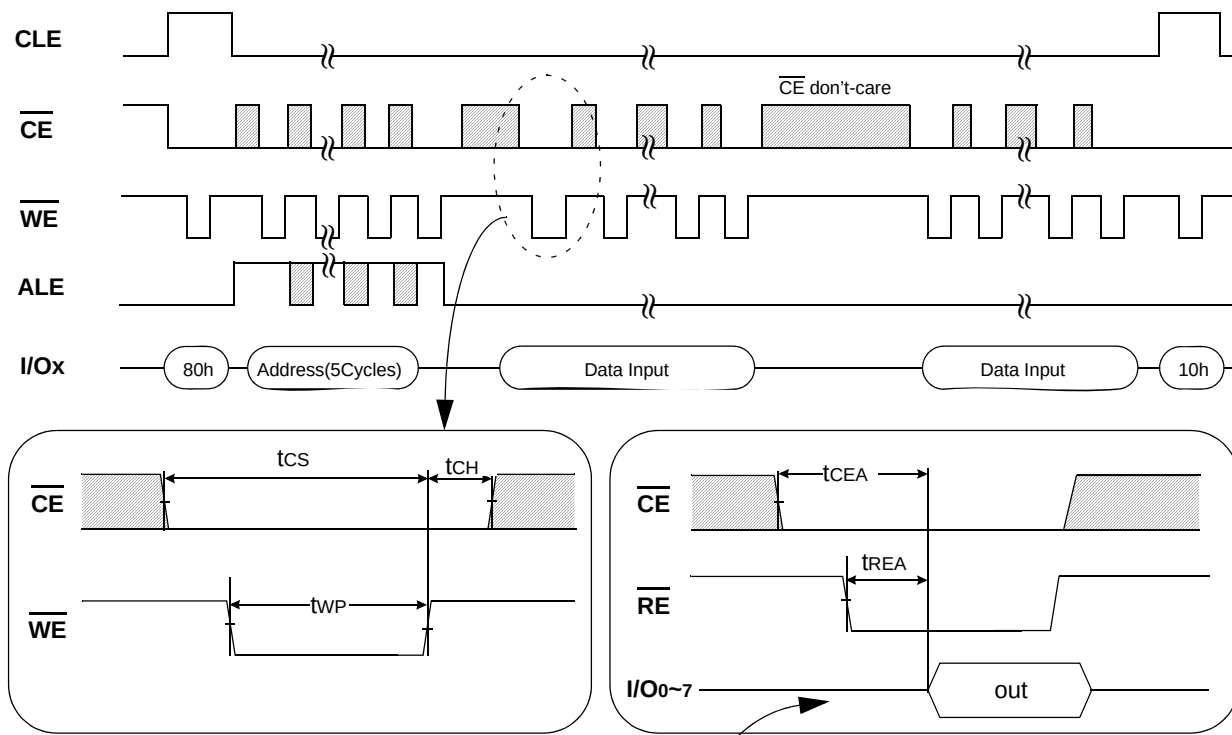
3.4 Addressing for Program Operation

Within a block, the pages must be programmed consecutively from the LSB(least significant bit) page of the block to the MSB(most significant bit) pages of the block. Random page address programming is prohibited. In this case, the definition of LSB page is the LSB among the pages to be programmed. Therefore, LSB doesn't need to be page 0.

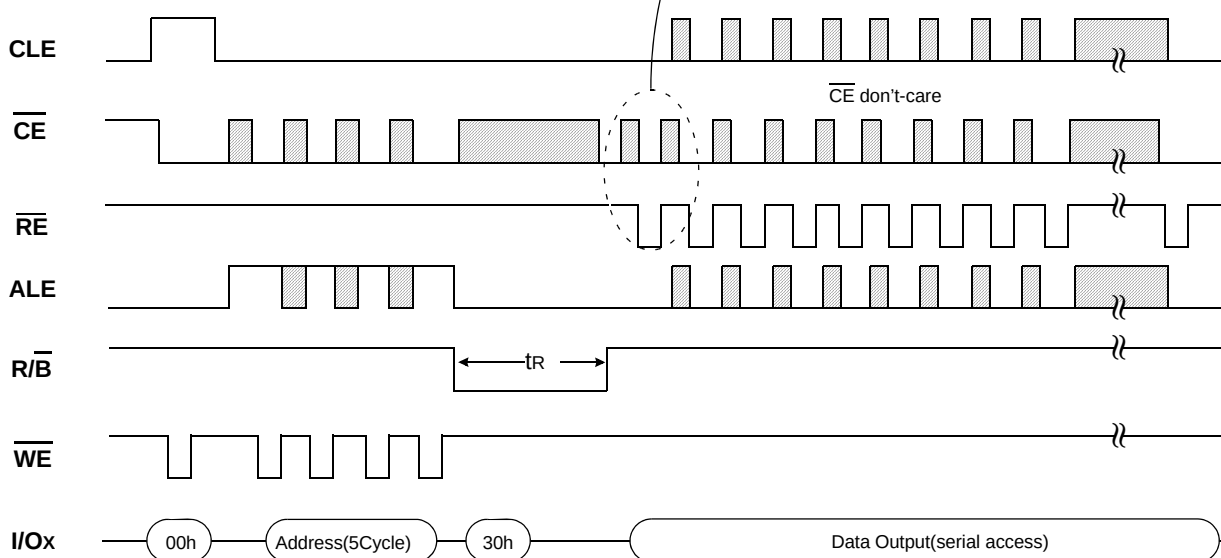


4.0 SYSTEM INTERFACE USING $\overline{\text{CE}}$ DON'T-CARE.

For an easier system interface, $\overline{\text{CE}}$ may be inactive during the data-loading or serial access as shown below. The internal 2,112byte data registers are utilized as separate buffers for this operation and the system design gets more flexible. In addition, for voice or audio applications which use slow cycle time on the order of μ -seconds, de-activating $\overline{\text{CE}}$ during the data-loading and serial access would provide significant savings in power consumption.



[Figure 2] Program Operation with $\overline{\text{CE}}$ don't-care.

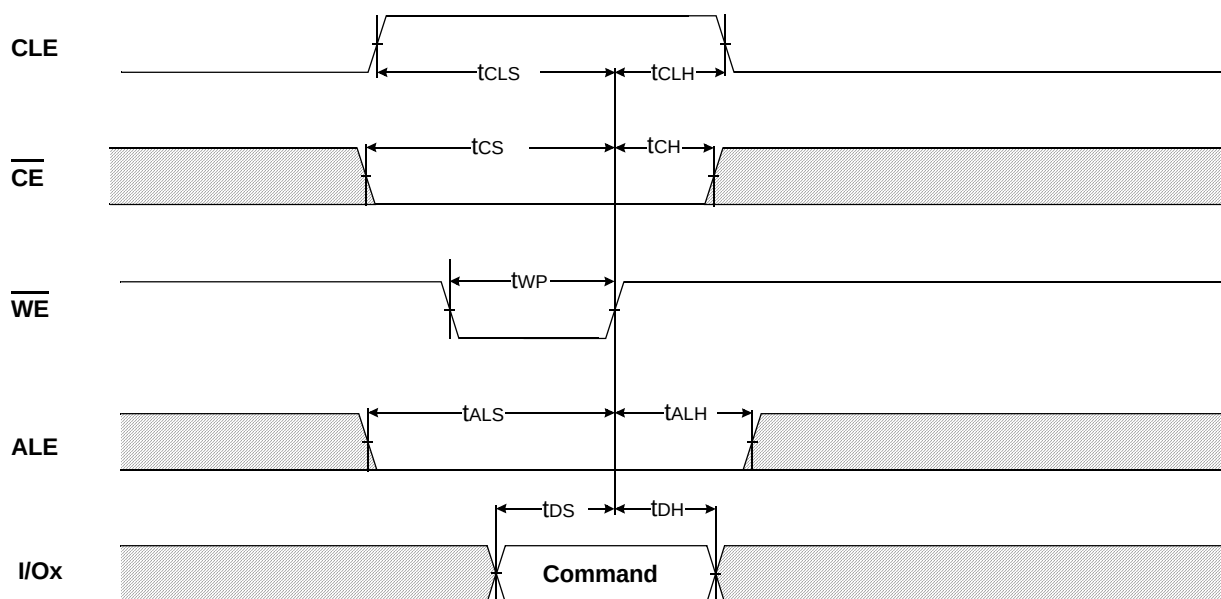


[Figure 3] Read Operation with $\overline{\text{CE}}$ don't-care.

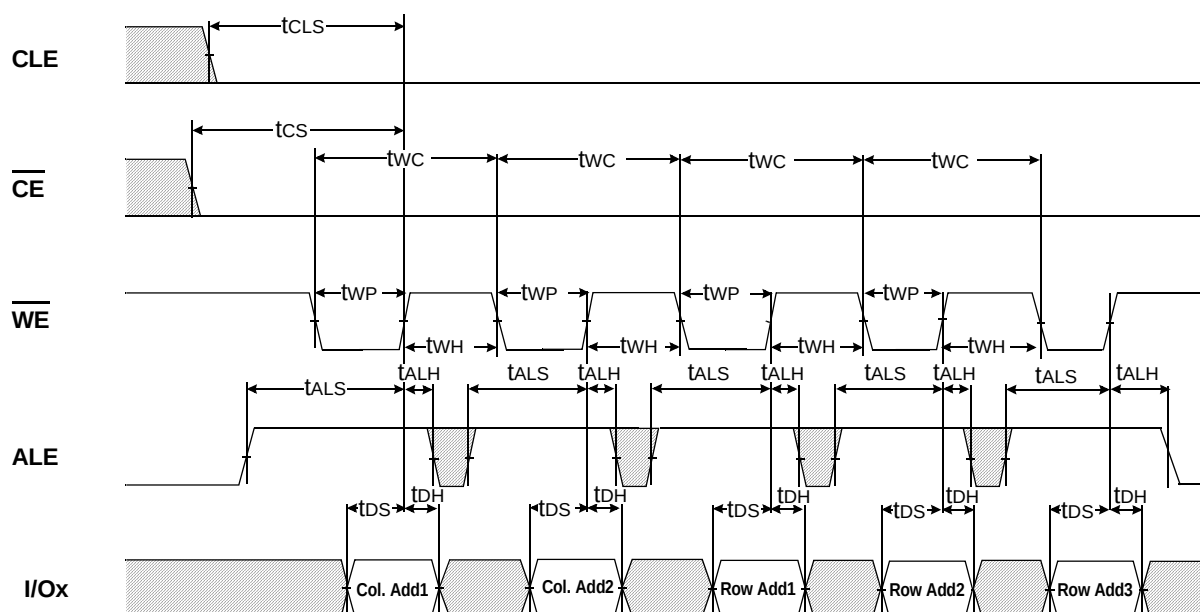
NOTE :

Device	I/O	DATA	ADDRESS				
	I/Ox	Data In/Out	Col. Add1	Col. Add2	Row Add1	Row Add2	Row Add3
K9F2G08U0C	I/O 0 ~ I/O 7	2,112byte	A0~A7	A8~A11	A12~A19	A20~A27	A28

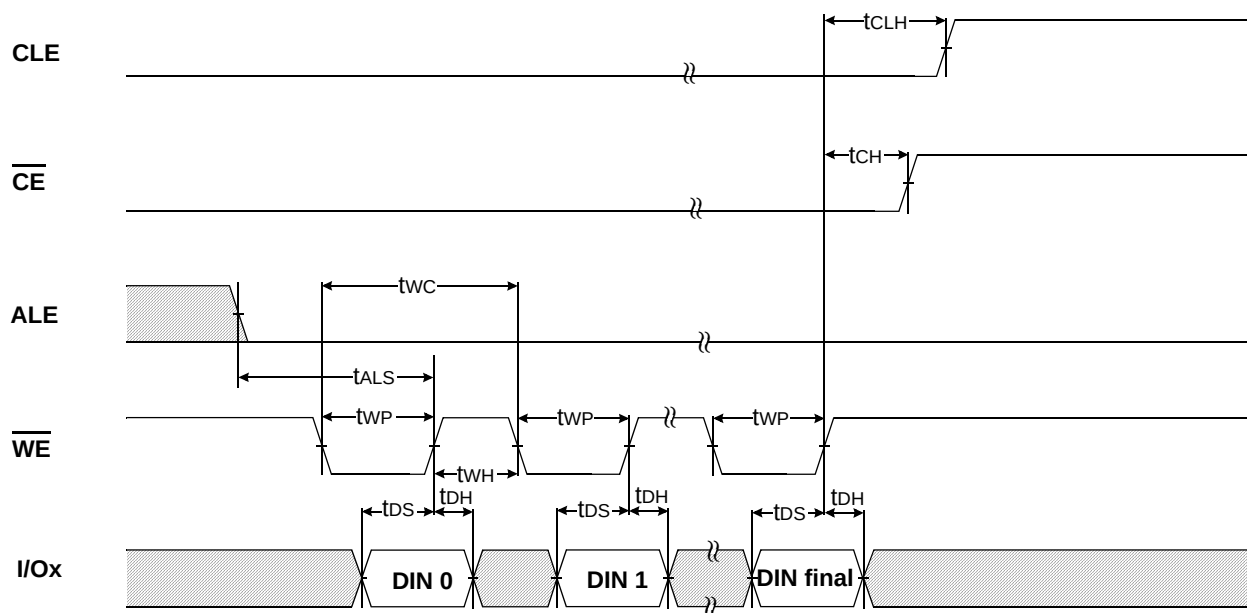
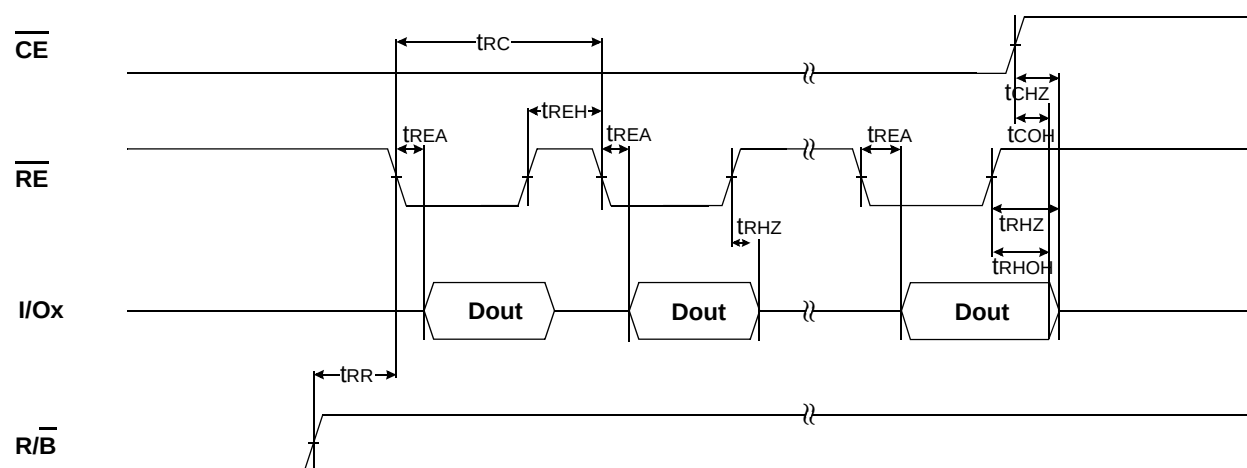
4.1 Command Latch Cycle



4.2 Address Latch Cycle

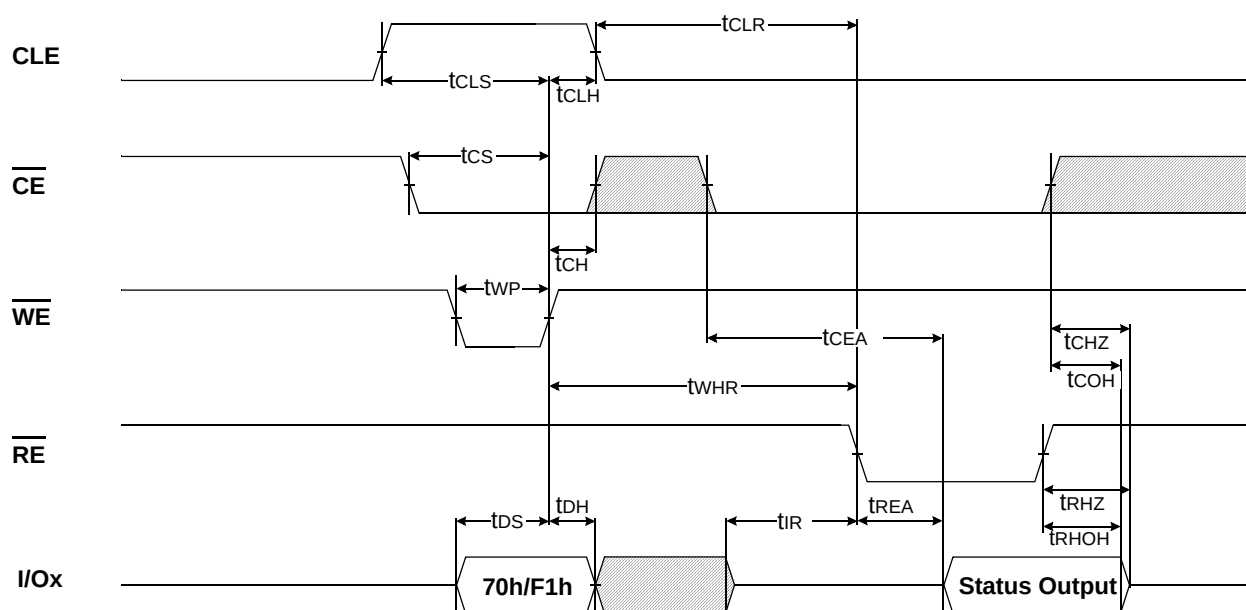


4.3 Input Data Latch Cycle

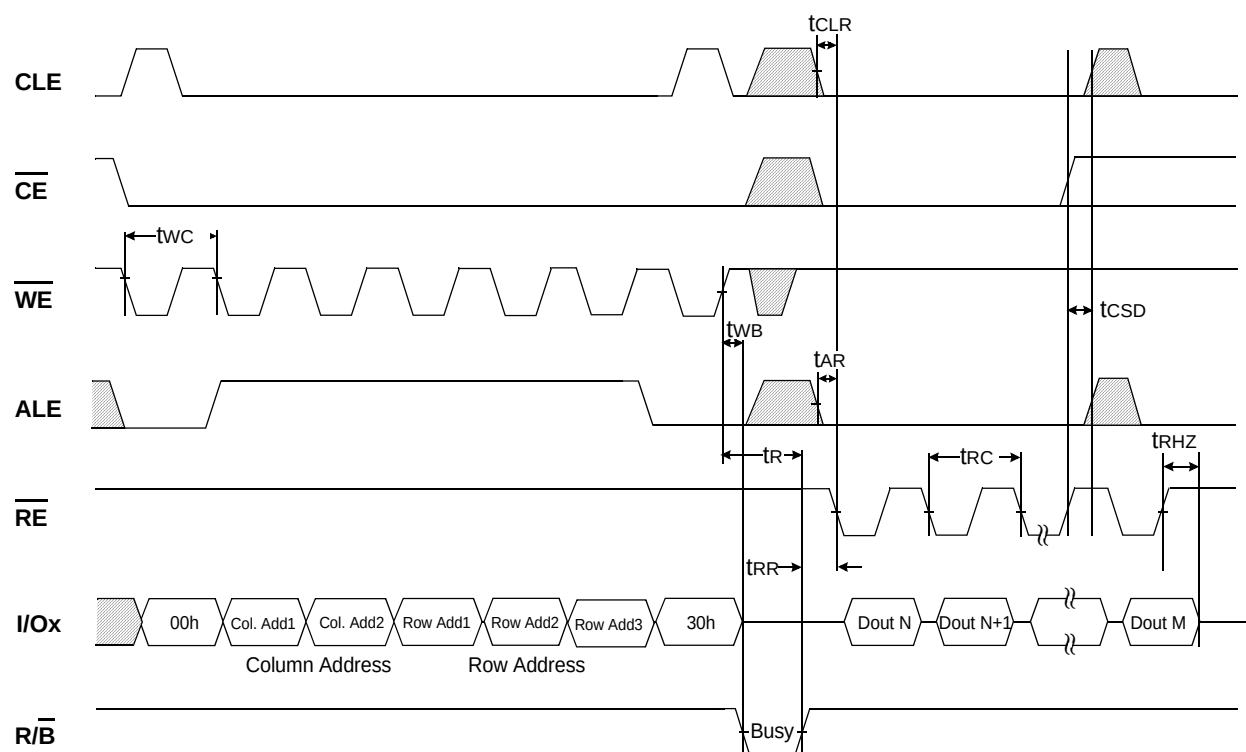
4.4* Serial Access Cycle after Read (CLE=L, $\overline{WE}$ =H, ALE=L)

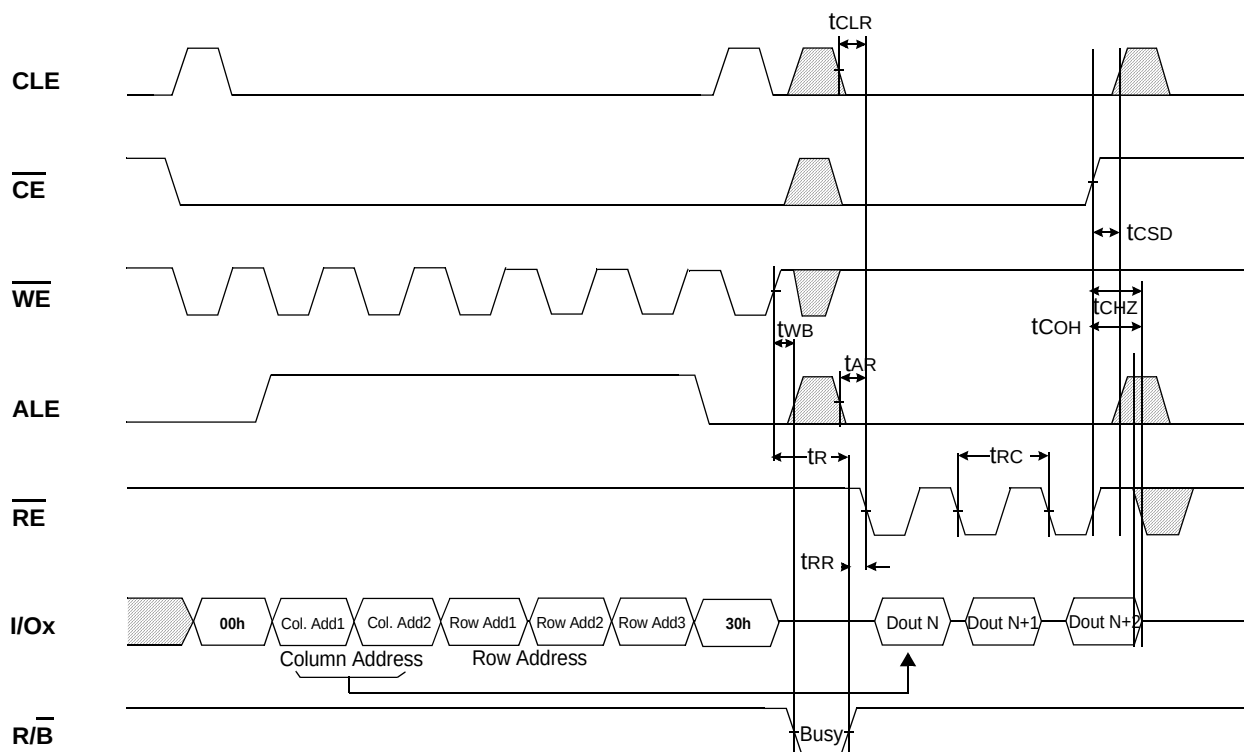
NOTE :
 Transition is measured at $\pm 200\text{mV}$ from steady state voltage with load.
 This parameter is sampled and not 100% tested.
 tRHOH starts to be valid when frequency is lower than 33MHz.

4.5 Status Read Cycle

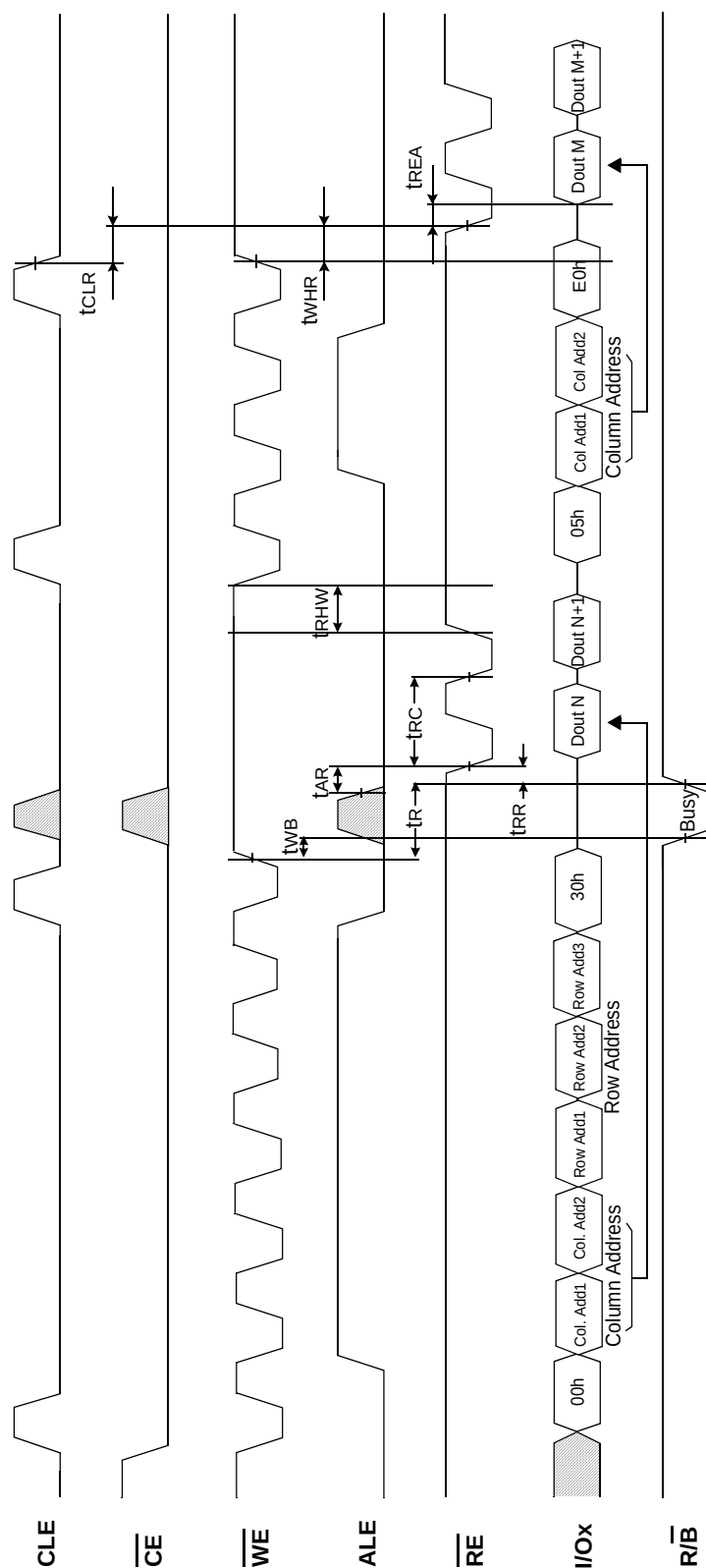


4.6 Read Operation

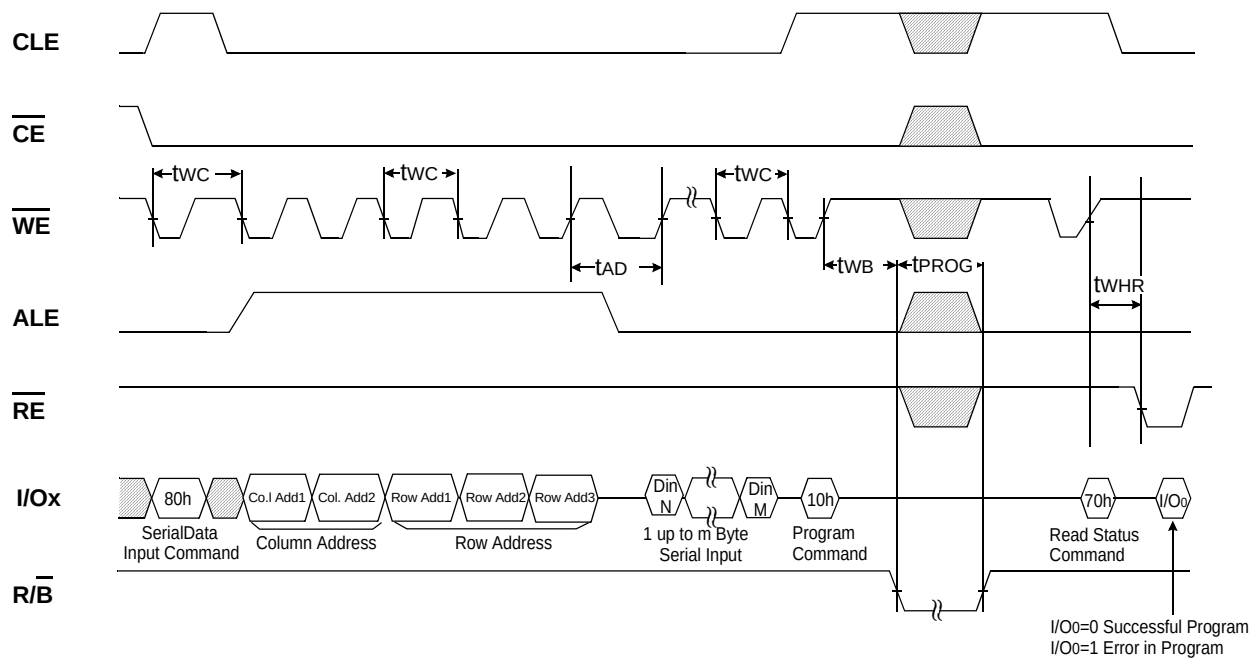


4.7 Read Operation(Intercepted by $\overline{\text{CE}}$)

4.8 Random Data Output In a Page

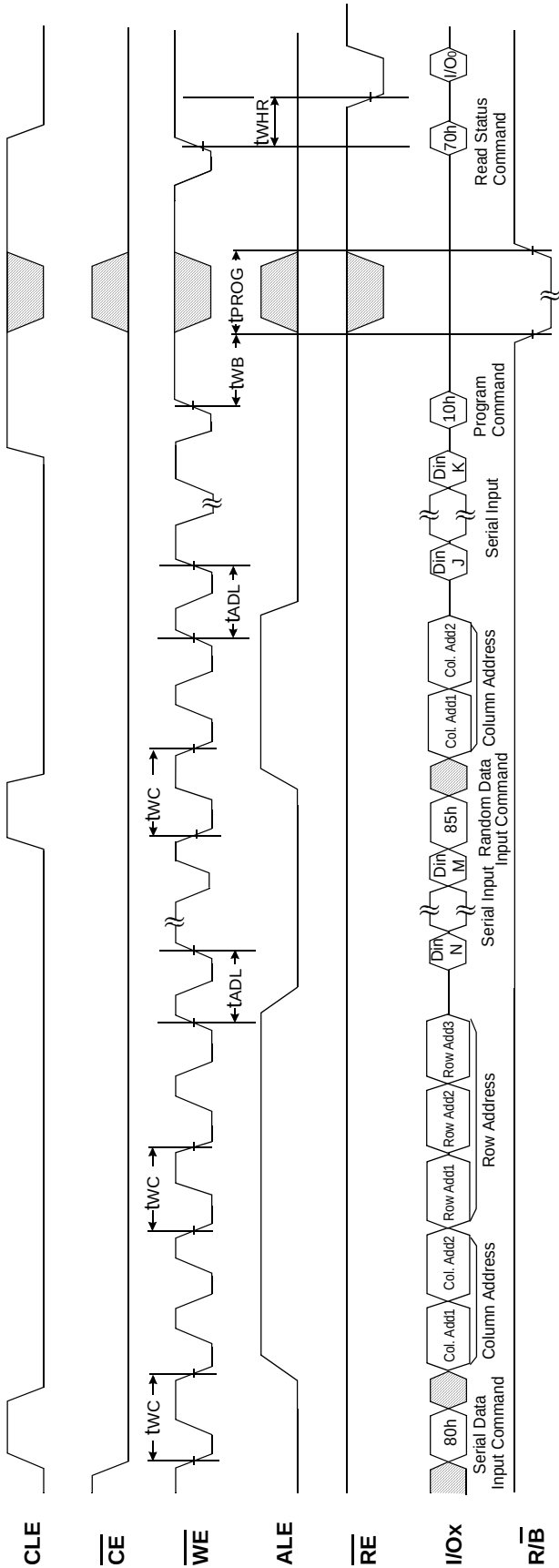


4.9 Page Program Operation



NOTE :
 t_{ADL} is the time from the $\overline{WE}$ rising edge of final address cycle to the $\overline{WE}$ rising edge of first data cycle.

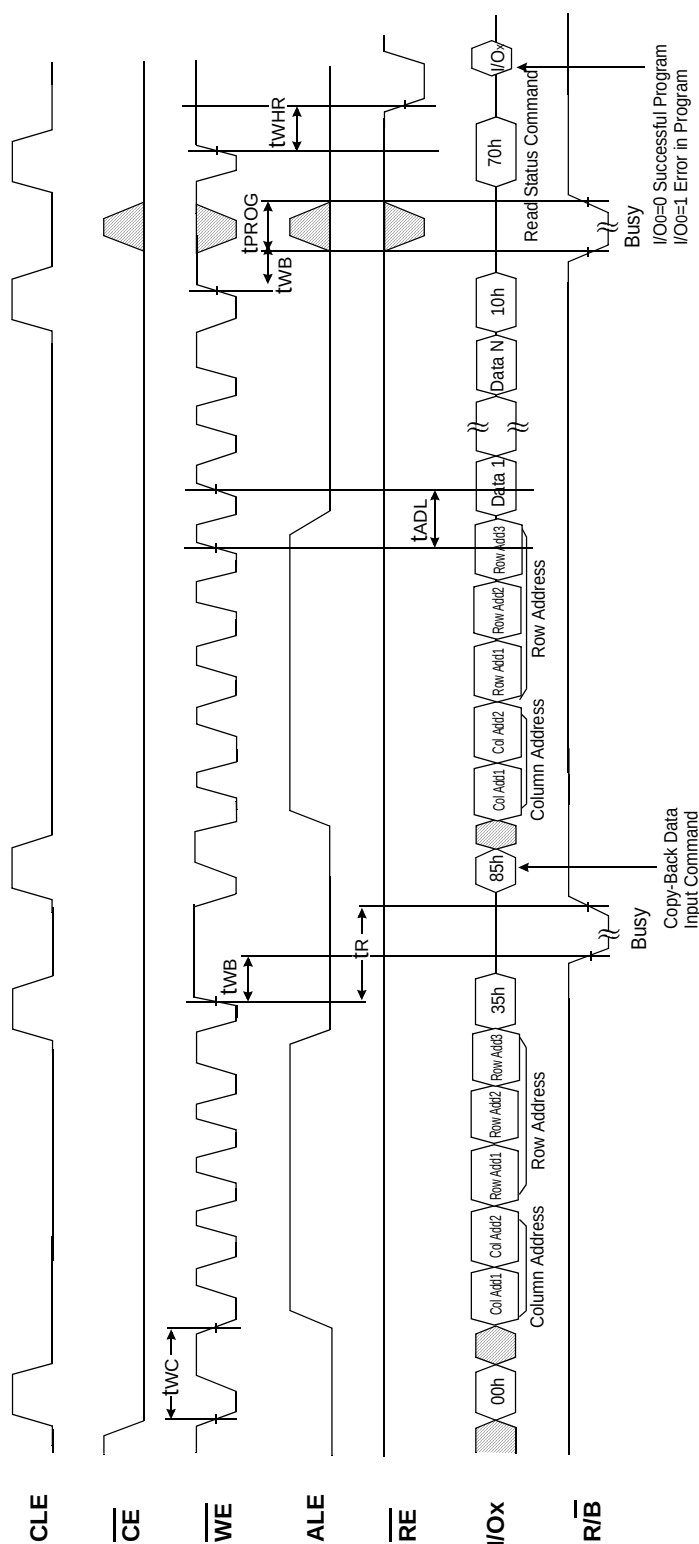
4.10 Page Program Operation with Random Data Input



NOTE :
tADL is h time from the WE rising edge of final address cycle to the WE rising edge of first data cycle.



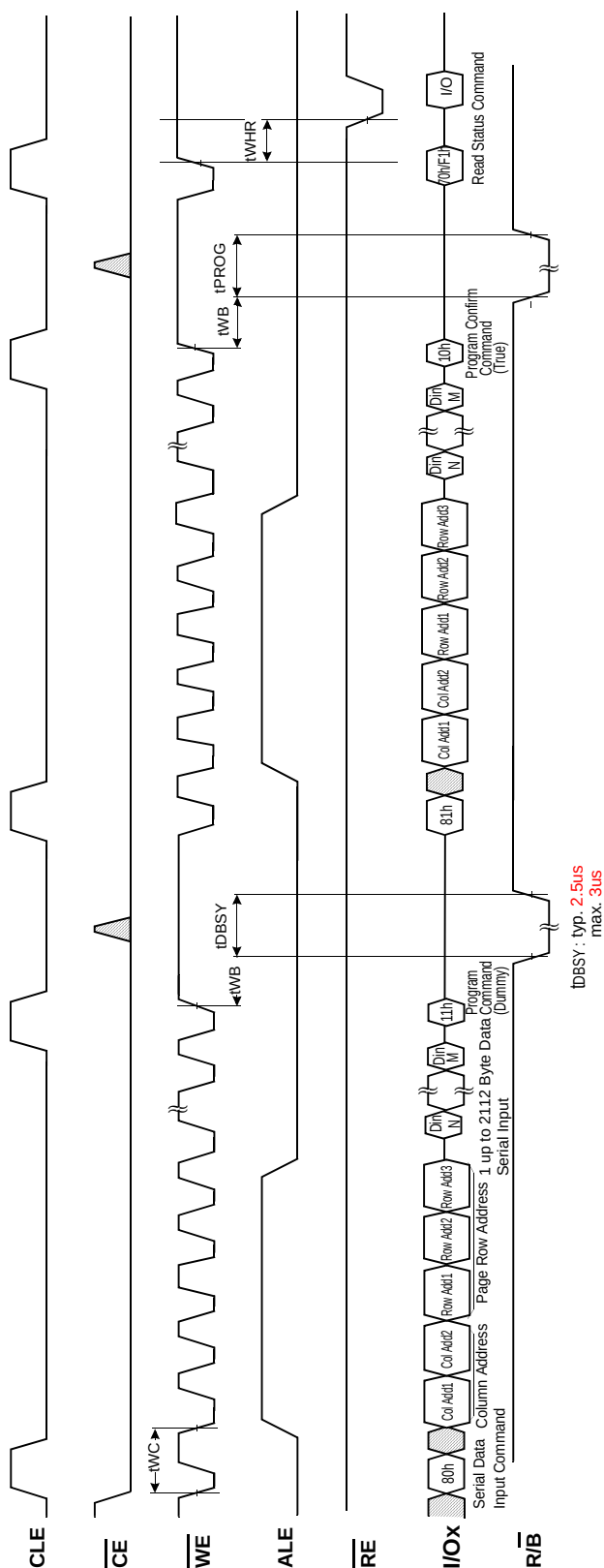
4.11 Copy-Back Program Operation With Random Data Input



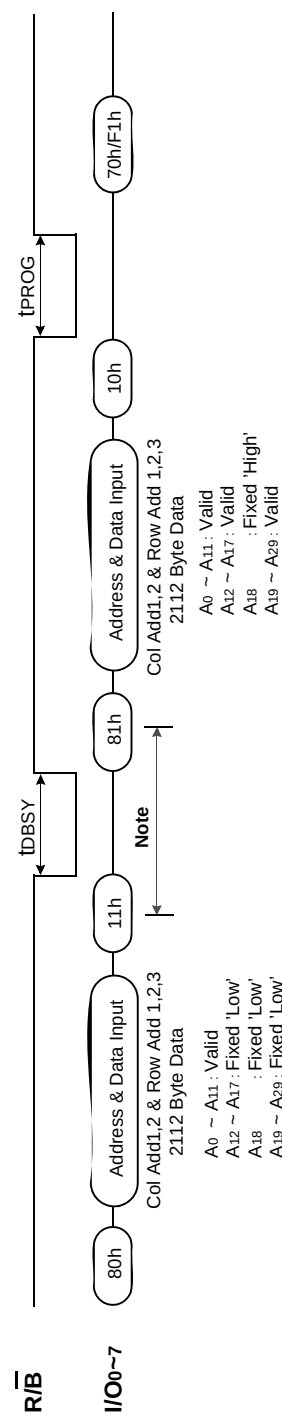
NOTE :

tADL is the time from the $\overline{WE}$ rising edge of final address cycle to the $\overline{WE}$ rising edge of first data cycle.

4.12 Two-Plane Page Program operation

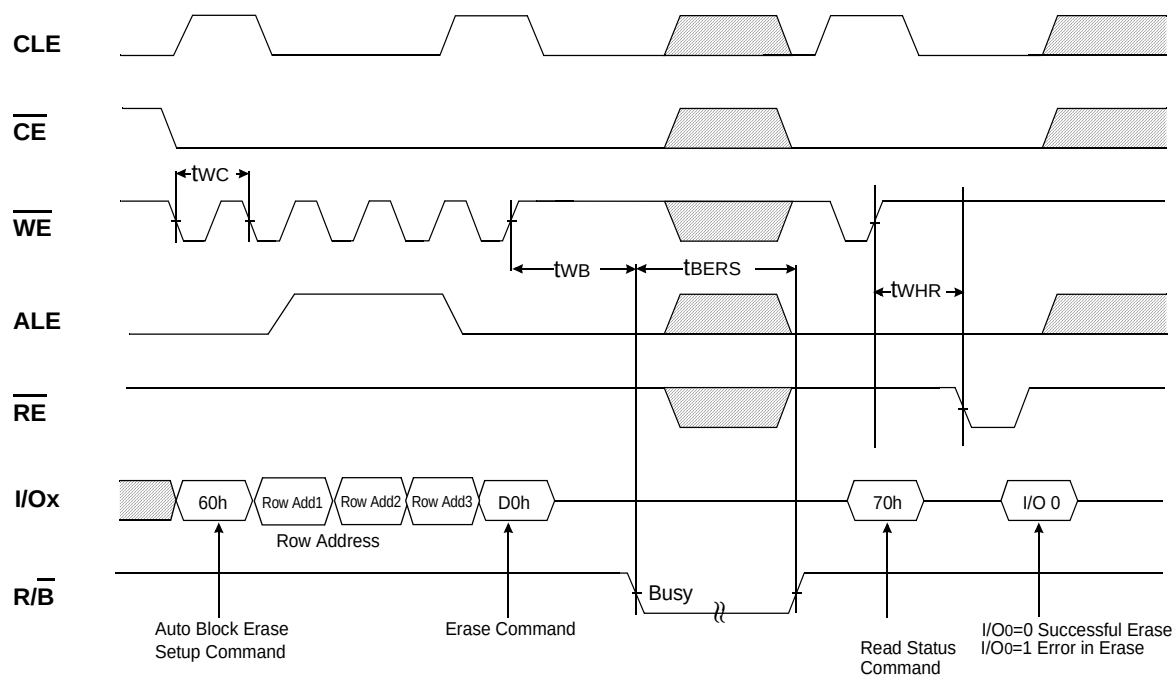


Ex.) Two-Plane Page Program

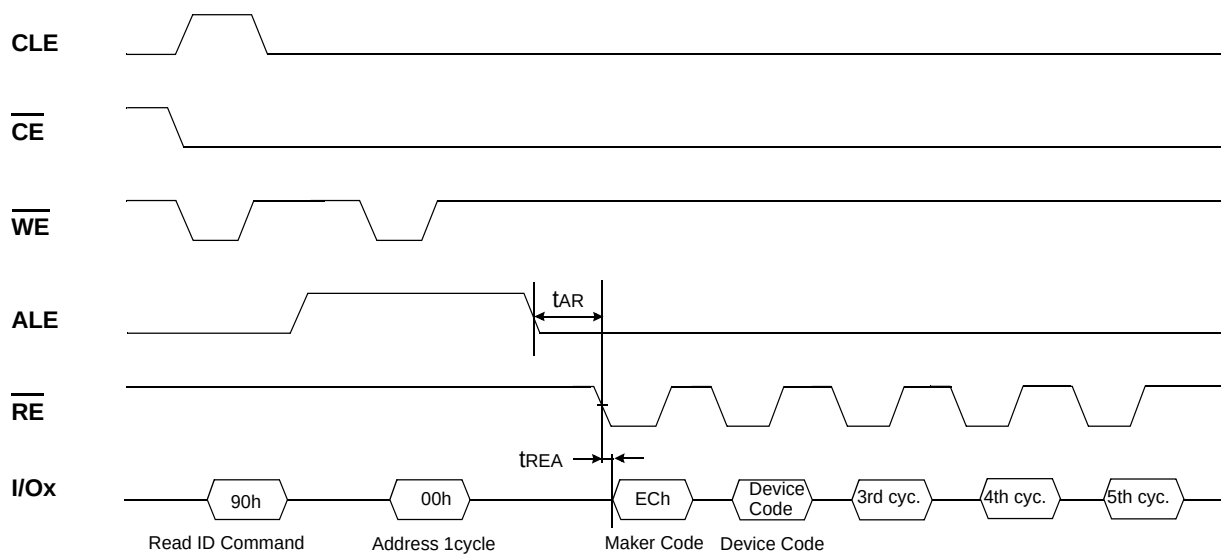


NOTE:
Any command between 11h and 81h is prohibited except 70h and FFh.

4.13 Block Erase Operation



4.14 Read ID Operation



Device	Device Code (2nd Cycle)	3rd Cycle	4th Cycle	5th Cycle
K9F2G08U0C	DAh	10h	15h	44h

ID Definition Table

	Description
1 st Byte	Maker Code
2 nd Byte	Device Code
3 rd Byte	Internal Chip Number, Cell Type, Number of Simultaneously Programmed Pages, Etc
4 th Byte	Page Size, Block Size, Redundant Area Size, Organization, Serial Access Minimum
5 th Byte	Plane Number, Plane Size

3rd ID Data

	Description	I/O7	I/O6	I/O5	I/O4	I/O3	I/O2	I/O1	I/O0
Internal Chip Number	1							0	0
	2							0	1
	4							1	0
	8							1	1
Cell Type	2 Level Cell					0	0		
	4 Level Cell					0	1		
	8 Level Cell					1	0		
	16 Level Cell					1	1		
Number of Simultaneously Programmed Pages	1			0	0				
	2			0	1				
	4			1	0				
	8			1	1				
Interleave Program Between multiple chips	Not Support		0						
	Support		1						
Cache Program	Not Support	0							
	Support	1							

4th ID Data

	Description	I/O7	I/O6	I/O5	I/O4	I/O3	I/O2	I/O1	I/O0
Page Size (w/o redundant area)	1KB							0	0
	2KB							0	1
	4KB							1	0
	8KB							1	1
Block Size (w/o redundant area)	64KB			0	0				
	128KB			0	1				
	256KB			1	0				
	512KB			1	1				
Redundant Area Size (byte/512byte)	8						0		
	16						1		
Organization	x8		0						
	x16		1						
Serial Access Minimum	50ns/30ns	0				0			
	25ns	1				0			
	Reserved	0				1			
	Reserved	1				1			

5th ID Data

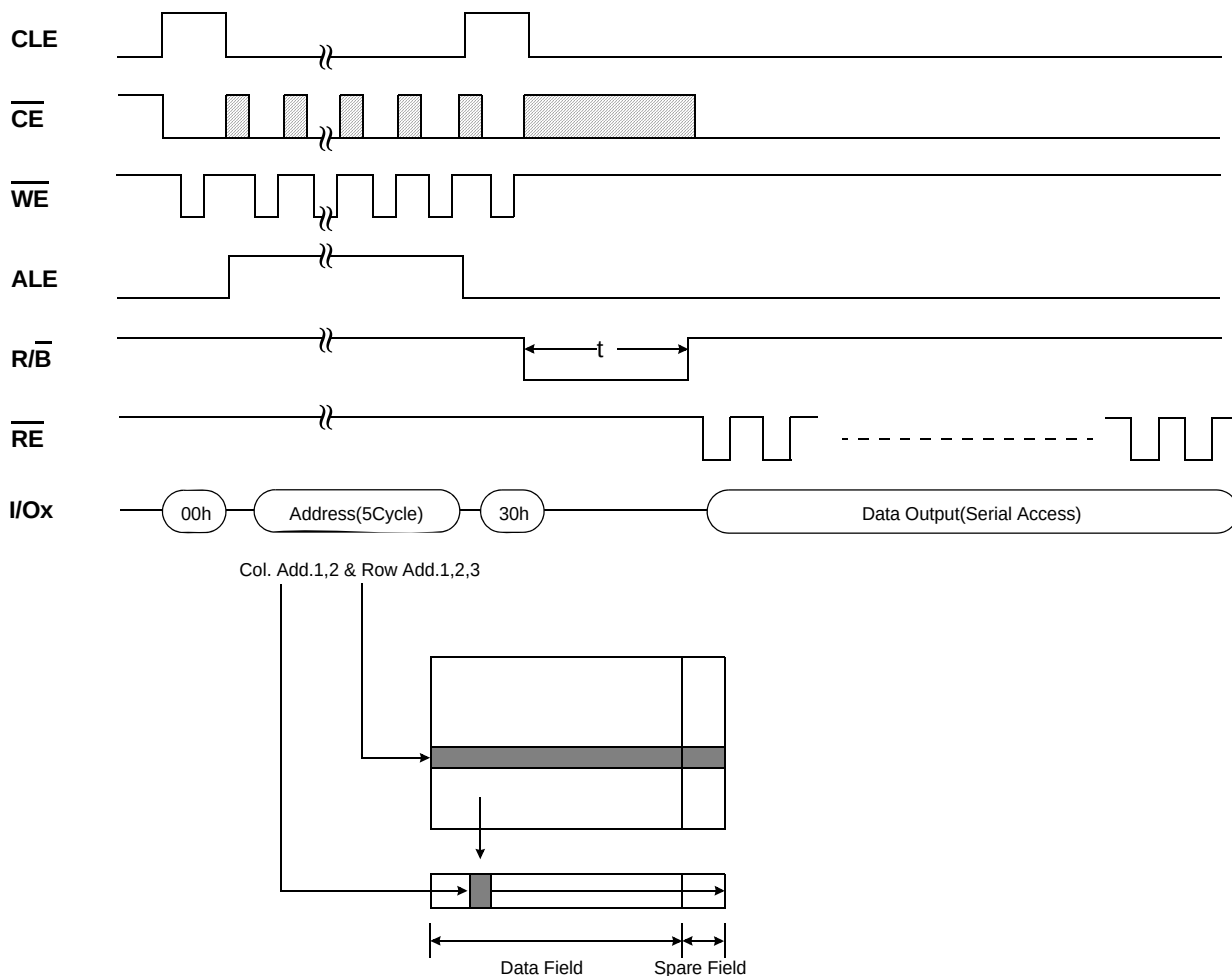
	Description	I/O7	I/O6	I/O5	I/O4	I/O3	I/O2	I/O1	I/O0
Plane Number	1					0	0		
	2					0	1		
	4					1	0		
	8					1	1		
Plane Size (w/o redundant Area)	64Mb		0	0	0				
	128Mb		0	0	1				
	256Mb		0	1	0				
	512Mb		0	1	1				
	1Gb		1	0	0				
	2Gb		1	0	1				
	4Gb		1	1	0				
	8Gb		1	1	1				
Reserved		0						0	0

5.0 DEVICE OPERATION

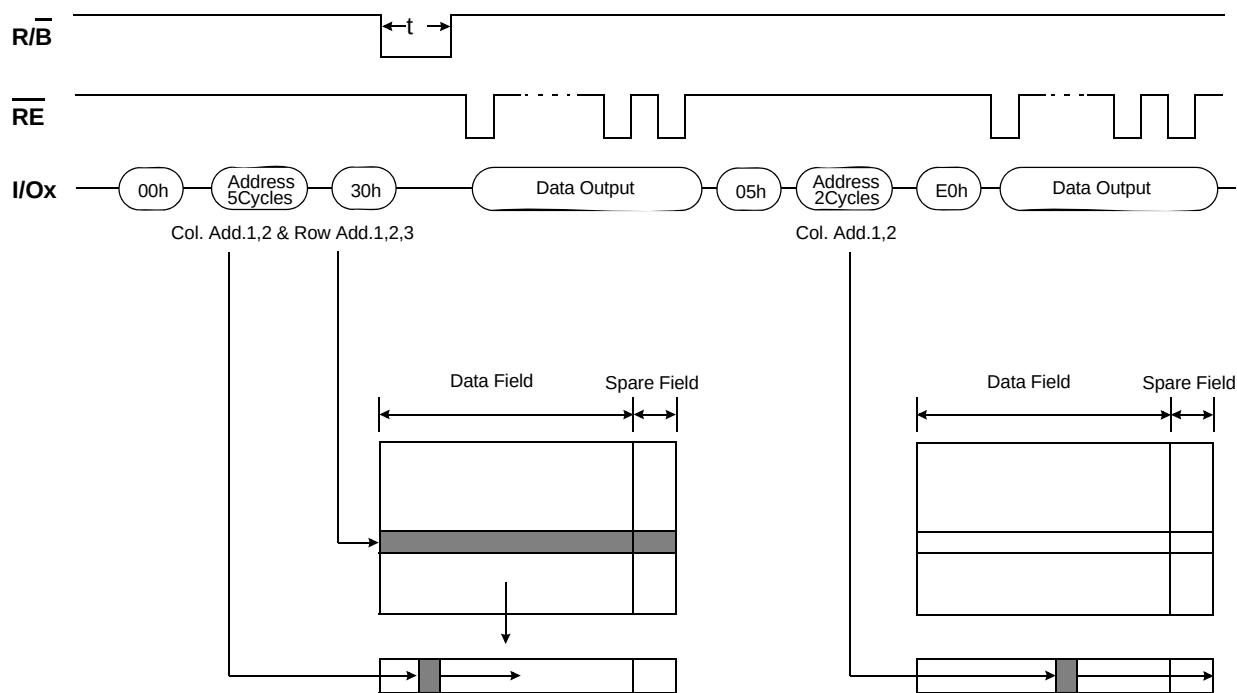
5.1 Page Read

Page read is initiated by writing 00h-30h to the command register along with five address cycles. After initial power up, 00h command is latched. Therefore only five address cycles and 30h command initiates that operation after initial power up. The 2,112 bytes of data within the selected page are transferred to the data registers in less than **40 μ s**(t_R). The system controller can detect the completion of this data transfer(t_R) by analyzing the output of R/B pin. Once the data in a page is loaded into the data registers, they may be read out in **25ns** cycle time by sequentially pulsing $\overline{\text{RE}}$. The repetitive high to low transitions of the $\overline{\text{RE}}$ clock make the device output the data starting from the selected column address up to the last column address.

The device may output random data in a page instead of the consecutive sequential data by writing random data output command. The column address of next data, which is going to be out, may be changed to the address which follows random data output command. Random data output can be operated multiple times regardless of how many times it is done in a page.



[Figure 4] Read Operation

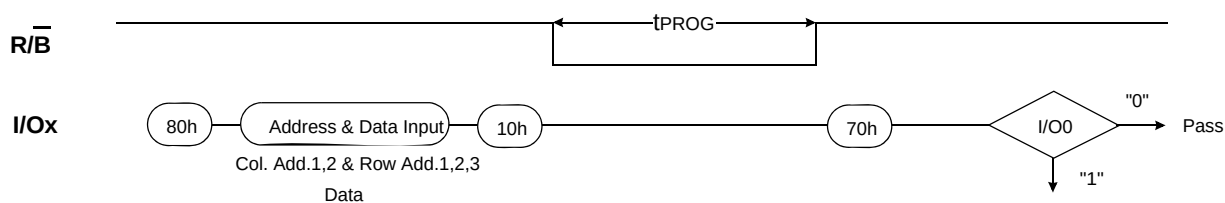


[Figure 5] Random Data Output In a Page

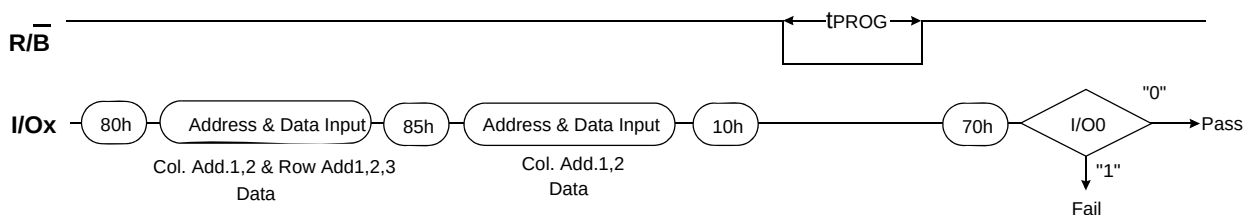
5.2 Page Program

The device is programmed basically on a page basis, but it does allow multiple partial page programming of a word or consecutive bytes up to 2,112, in a single page program cycle. The number of consecutive partial page programming operation within the same page without an intervening erase operation must not exceed 4 times for a single page. The addressing should be done in sequential order in a block. A page program cycle consists of a serial data loading period in which up to 2,112bytes of data may be loaded into the data register, followed by a non-volatile programming period where the loaded data is programmed into the appropriate cell.

The serial data loading period begins by inputting the Serial Data Input command(80h), followed by the five cycle address inputs and then serial data loading. The words other than those to be programmed do not need to be loaded. The device supports random data input in a page. The column address for the next data, which will be entered, may be changed to the address which follows random data input command(85h). Random data input may be operated multiple times regardless of how many times it is done in a page. The Page Program confirm command(10h) initiates the programming process. Writing 10h alone without previously entering the serial data will not initiate the programming process. The internal write state controller automatically executes the algorithms and timings necessary for program and verify, thereby freeing the system controller for other tasks. Once the program process starts, the Read Status Register command may be entered to read the status register. The system controller can detect the completion of a program cycle by monitoring the R/B output, or the Status bit(I/O 6) of the Status Register. Only the Read Status command and Reset command are valid while programming is in progress. When the Page Program is complete, the Write Status Bit(I/O 0) may be checked(Figure 6). The internal write verify detects only errors for "1"s that are not successfully programmed to "0"s. The command register remains in Read Status command mode until another valid command is written to the command register.



[Figure 6] Program & Read Status Operation

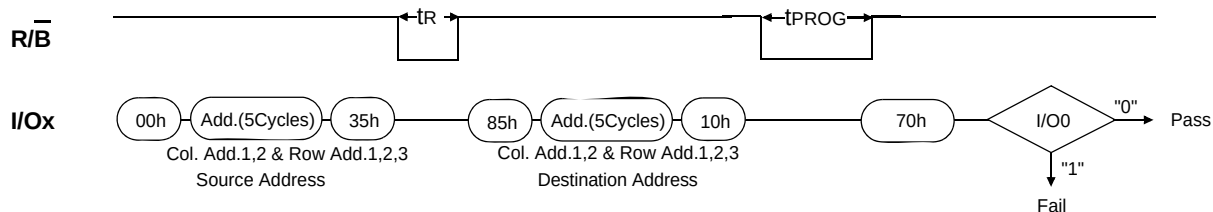


[Figure 7] Random Data Input In a Page

5.3 Copy-back Program

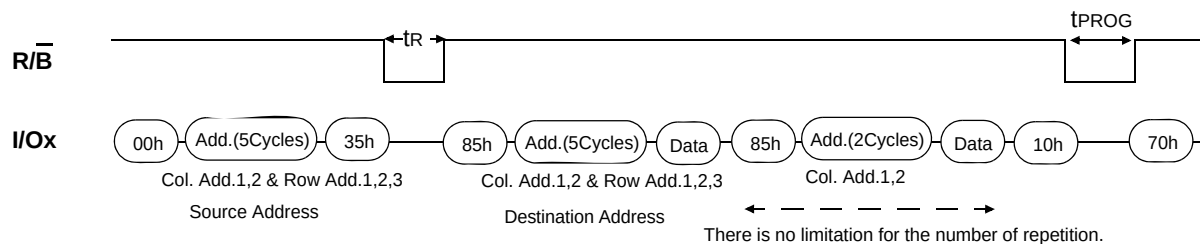
Copy-Back program with Read for Copy-Back is configured to quickly and efficiently rewrite data stored in one page without data re-loading when the bit error is not in data stored. Since the time-consuming re-loading cycles are removed, the system performance is improved. The benefit is especially obvious when a portion of a block is updated and the rest of the block also needs to be copied to the newly assigned free block. Copy-Back operation is a sequential execution of Read for Copy-Back and of copy-back program with the destination page address. A read operation with "35h" command and the address of the source page moves the whole 2,112-byte data into the internal data buffer. A bit error is checked by sequential reading the data output. In the case where there is no bit error, the data do not need to be reloaded. Therefore Copy-Back program operation is initiated by issuing Page-Copy Data-Input command (85h) with destination page address. Actual programming operation begins after Program Confirm command (10h) is issued. Once the program process starts, the Read Status Register command (70h) may be entered to read the status register. The system controller can detect the completion of a program cycle by monitoring the $\overline{R}/\overline{B}$ output, or the Status bit(I/O 6) of the Status Register. When the Copy-Back Program is complete, the Write Status Bit(I/O 0) may be checked(Figure 8 & Figure 9). The command register remains in Read Status command mode until another valid command is written to the command register.

During copy-back program, data modification is possible using random data input command (85h) as shown in Figure 9.



NOTE :
Copy-Back Program operation is allowed only within the same memory plane.

[Figure 8] Page Copy-Back Program Operation



[Figure 9] Page Copy-Back Program Operation with Random Data Input

5.4 Read Status

The device contains a Status Register which may be read to find out whether program or erase operation is completed, and whether the program or erase operation is completed successfully. After writing 70h/F1h command to the command register, a read cycle outputs the content of the Status Register to the I/O pins on the falling edge of $\overline{CE}$ or $\overline{RE}$, whichever occurs last. This two line control allows the system to poll the progress of each device in multiple memory connections even when $R/\overline{B}$ pins are common-wired. $\overline{RE}$ or $\overline{CE}$ does not need to be toggled for updated status. Refer to Table 2 for specific Status Register definitions and Table 3 for specific F1h Status Register definitions. The command register remains in Status Read mode until further commands are issued to it. Therefore, if the status register is read during a random read cycle, the read command(00h) should be given before starting read cycles.

[Table 2] Read Status Register Definition for 70h Command

I/O	Page Program	Block Erase	Read	Definition
I/O 0	Pass/Fail	Pass/Fail	Not use	Pass : "0" Fail : "1"
I/O 1	Not use	Not use	Not use	Don't -cared
I/O 2	Not use	Not use	Not use	Don't -cared
I/O 3	Not Use	Not Use	Not Use	Don't -cared
I/O 4	Not Use	Not Use	Not Use	Don't -cared
I/O 5	Not Use	Not Use	Not Use	Don't -cared
I/O 6	Ready/Busy	Ready/Busy	Ready/Busy	Busy : "0" Ready : "1"
I/O 7	Write Protect	Write Protect	Write Protect	Protected : "0" Not Protected : "1" "1"otected

NOTE :

I/Os defined 'Not use' are recommended to be masked out when Read Status is being executed.

[Table 3] Read Status 2 Register Definition for F1h Command

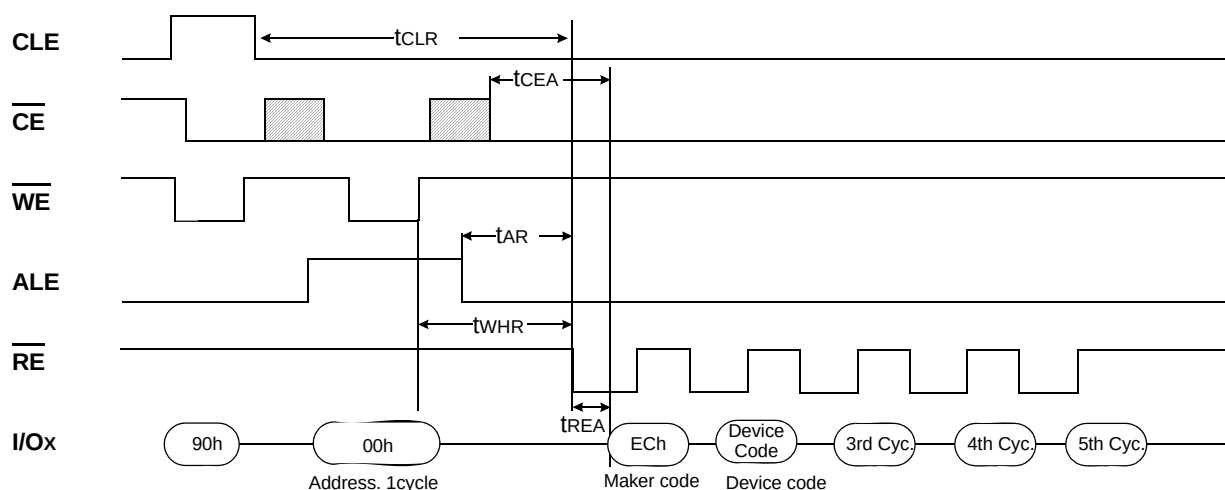
I/O No.	Page Program	Block Erase	Read	Definition
I/O 0	Chip Pass/Fail	Chip Pass/Fail	Not use	Pass : "0" Fail : "1"
I/O 1	Plane0 Pass/Fail	Plane0 Pass/Fail	Not use	Pass : "0" Fail : "1"
I/O 2	Plane1 Pass/Fail	Plane1 Pass/Fail	Not use	Pass : "0" Fail : "1"
I/O 3	Not Use	Not Use	Not Use	Don't -cared
I/O 4	Not Use	Not Use	Not Use	Don't -cared
I/O 5	Not Use	Not Use	Not Use	Don't -cared
I/O 6	Ready/Busy	Ready/Busy	Ready/Busy	Busy : "0" Ready : "1"
I/O 7	Write Protect	Write Protect	Write Protect	Protected : "0" Not Protected : "1" "1"otected

NOTE :

I/Os defined 'Not use' are recommended to be masked out when Read Status is being executed.

5.5 Read ID

The device contains a product identification mode, initiated by writing 90h to the command register, followed by an address input of 00h. Five read cycles sequentially output the manufacturer code (ECh), and the device code and 3rd, 4th, 5th cycle ID respectively. The command register remains in Read ID mode until further commands are issued to it. Figure 10 shows the operation sequence.



[Figure 10] Read ID Operation

Device	Device Code (2nd Cycle)	3rd Cycle	4th Cycle	5th Cycle
K9F2G08U0C	DAh	10h	15h	44h

5.6 Reset

The device offers a reset feature, executed by writing FFh to the command register. When the device is in Busy state during random read, program or erase mode, the reset operation will abort these operations. The contents of memory cells being altered are no longer valid, as the data will be partially programmed or erased. The command register is cleared to wait for the next command, and the Status Register is cleared to value C0h when $\overline{WP}$ is high. If the device is already in reset state a new reset command will be accepted by the command register. The R/B pin changes to low for tRST after the Reset command is written. Refer to Figure 11 below.

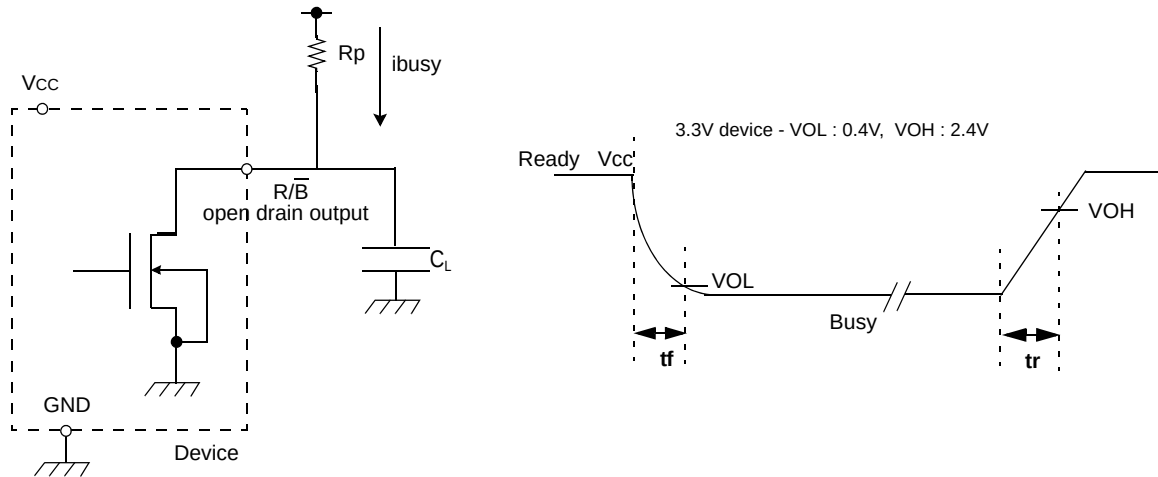


[Figure 11] RESET Operation

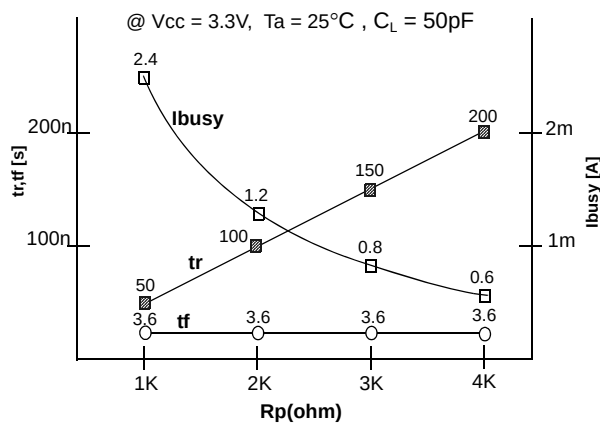
	After Power-up	After Reset
Operation mode Mode	00h Command is latched	Waiting for next command

5.7 READY/ $\overline{\text{BUSY}}$

The device has a $\overline{\text{R/B}}$ output that provides a hardware method of indicating the completion of a page program, erase and random read completion. The $\overline{\text{R/B}}$ pin is normally high but transitions to low after program or erase command is written to the command register or random read is started after address loading. It returns to high when the internal controller has finished the operation. The pin is an open-drain driver thereby allowing two or more $\overline{\text{R/B}}$ outputs to be Or-tied. Because pull-up resistor value is related to $t_r(\overline{\text{R/B}})$ and current drain during busy(i_{busy}), an appropriate value can be obtained with the following reference chart(Figure 12). Its value can be determined by the following guidance.



[Figure 12] R_p vs t_r, t_f & R_p vs i_{busy}



R_p value guidance

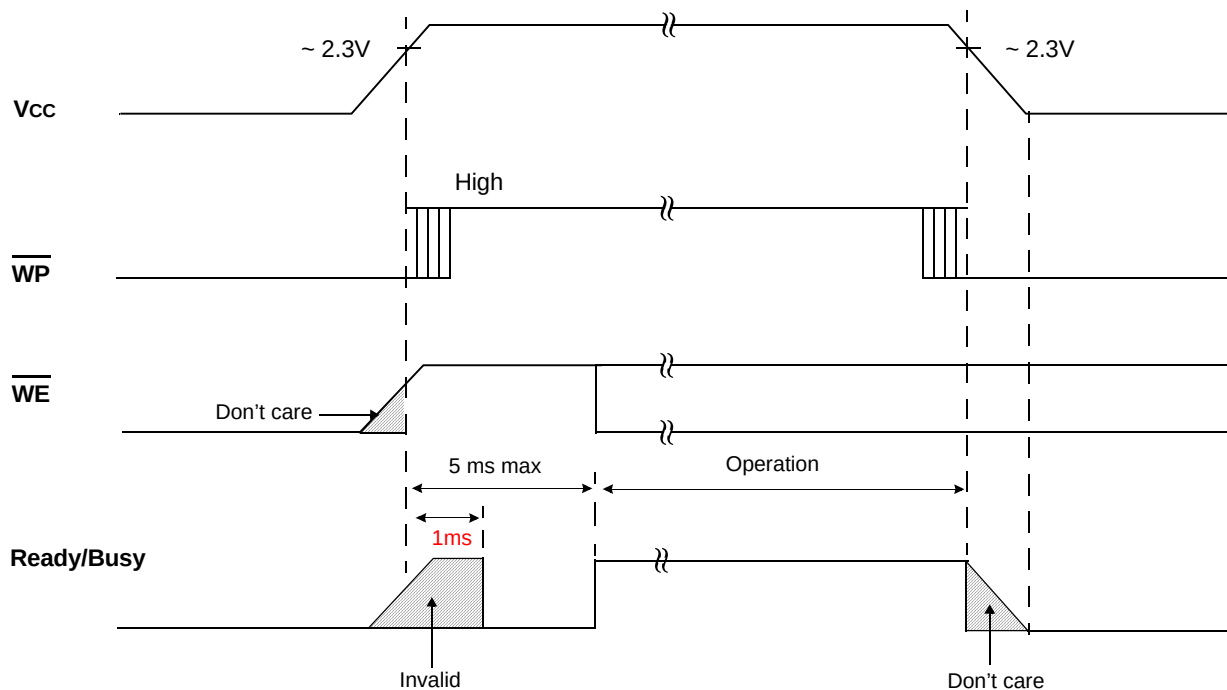
$$R_{p(\text{min}, 3.3\text{V part})} = \frac{V_{\text{cc}(\text{Max.})} - V_{\text{OL}(\text{Max.})}}{I_{\text{OL}} + \sum I_L} = \frac{3.2\text{V}}{8\text{mA} + \sum I_L}$$

where I_L is the sum of the input currents of all devices tied to the $\overline{\text{R/B}}$ pin.

$R_{p(\text{max})}$ is determined by maximum permissible limit of t_r

6.0 DATA PROTECTION & POWER UP SEQUENCE

The device is designed to offer protection from any involuntary program/erase during power-transitions. An internal voltage detector disables all functions whenever V_{CC} is below about 2V(3.3V device). $\overline{WP}$ pin provides hardware protection and is recommended to be kept at V_{IL} during power-up and power-down. A recovery time of minimum **1ms** is required before internal circuit gets ready for any command sequences as shown in Figure 13. The two step command sequence for program/erase provides additional software protection.



[Figure 13] AC Waveforms for Power Transition

7.0 BACKWARD COMPATIBILITY INFORMATION

The below table shows key parameters which are different with previous product, so that the host could use make or modify its firmware without misunderstanding of compatibility. But the below table don't have all the difference with previous product, but only key parameters' changing which can be defined to have an effect on developing NAND firmware or hardware.

	Previous Generation Product	Current Generation Device
Part ID	K9F2G08U0B	K9F2G08U0C
Features & Operations	1. tR: 25us / tPROG(200us typ, 700us Max) tERS(1.5ms Typ, 10ms Max) 2. tRC/tWC: 25ns 3. 2 Plane Program: support 4. 2Plane Copy-back Program: Support 5. 2Plane Erase: Support 6. EDO: Support	1. tR: 40us / tPROG(250us typ, 750us Max) tERS(2ms Typ, 10ms Max) 2. tRC/tWC: 25ns 3. 2 Plane Program: support 4. 2Plane Copy-back Program: N/A 5. 2Plane Erase: N/A 6. EDO: N/A
AC & DC Parameters	1. ICC1 : 15mA(typ)/ 30mA(max) 2. ICC2 : 15mA(typ)/ 30mA(max) 3. ICC3 : 15mA(typ)/ 30mA(max)	1. ICC1 : 20mA(typ)/ 35mA(max) 2. ICC2 : 20mA(typ)/ 35mA(max) 3. ICC3 : 20mA(typ)/ 35mA(max)
Technical Notes		